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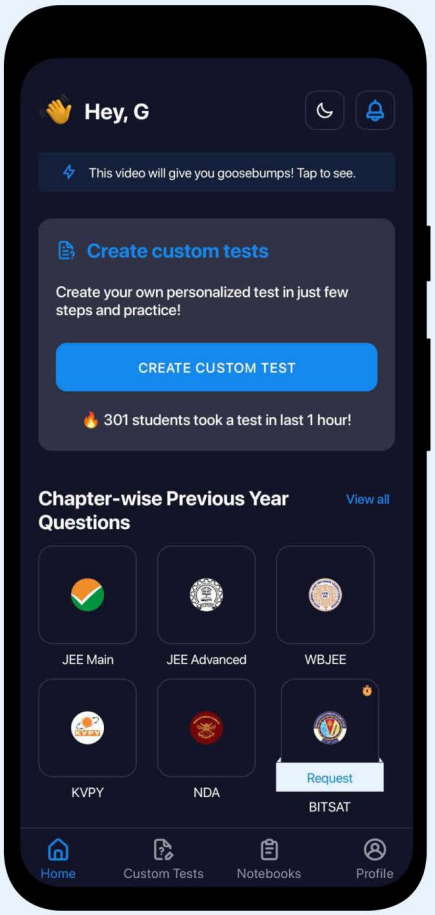

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PARABOLA

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PARABOLA

1. INTRODUCTION TO CONICS

1.1 Definition

Conic Section :— The locus of a point which moves such that its distance from a fixed point called focus bears a constant ratio **(called eccentricity denoted by the letter e)** to its distance from a fixed straight line called **directrix** is called a conic section.

The above conic section is a hyperbola or ellipse according as the constant ratio e is greater or less than unity.

In case the ratio e is equal to unity, then the locus of the moving point is called parabola. Hence we can define a parabola as follows :—

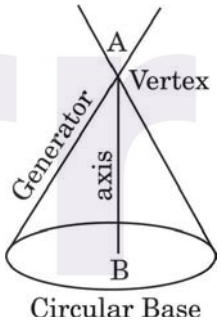
"A parabola is the locus of a point such that its distance from a fixed point called focus is equal to its distance from a fixed straight line called directrix.

1.2 JUSTIFICATION FOR THE NAME CONIC SECTION :

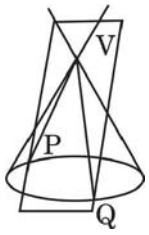
Parabola, ellipse and hyperbola are sections of a right circular cone by a plane and so they are called conic section. Circle and pair of straight lines are also section of a right circular cone by a plane and they are also called conic sections.

1.3 Section of a right circular cone by different planes :

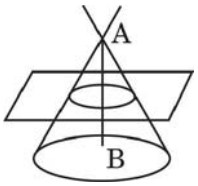
A right circular cone is as shown in the figure.



1. Section of a right circular cone by a plane passing through its vertex is a pair of straight lines passing through the vertex as shown in the figure.

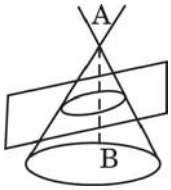


2. Section of a right circular cone by a plane parallel to its base is a circle as shown in the figure.

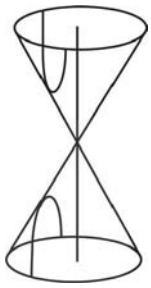


3. Section of a right circular cone by a plane parallel to a generator of the cone is a parabola as shown in the figure.

4. Section of a right circular cone by a plane not parallel to any generator of the cone and not perpendicular or parallel to the axis of the cone is an ellipse as shown in the figure.



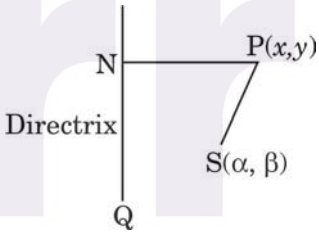
5. Section of a right circular cone by a plane parallel to the axis of the cone is a hyperbola as shown in the figure.



1.4 Equation of Conic from Focus-Directrix Property :

Let S(α, β) be the focus and QN the directrix whose equation is

$$A x + By + C = 0 \qquad ...(1)$$



Let P (x, y) be any point on the conic section.

From P draw PN ⊥ QN.

If e be the eccentricity of the conic, then by definition

$$\frac{PS}{PN} = e \qquad ...(2)$$

From (2), $$PS^2 = e^2PN^2 \qquad ...(3)$$

But $PS^2 = (x - \alpha)^2 + (y - \beta)^2$ and $PN^2 = \left[\frac{A x + B y + C}{\sqrt{A^2 + B^2}}\right]^2$

∴ From (3), $(x - \alpha)^2 + (y - \beta)^2 = e^2 \cdot \left(\frac{A x + B y + C}{\sqrt{A^2 + B^2}}\right)^2$

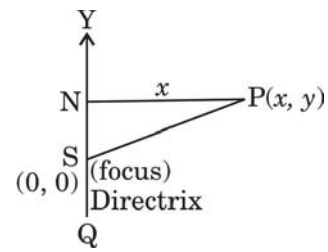
which, on simplification takes the form

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \qquad ...(4)$$

where a, b, c, f, g and h are constants.(4) being the locus of P, is the equation of the conic.

Thus we see that the equation of a conic is an equation of second degree in x and y.

Note : 1. Let the focus S lie on the directrix QN and S be taken as the origin and QN be taken as the y–axis. Let P(x, y) be any point on the conic, then



$$\frac{PS}{PN} = e; \text{ or } \frac{\sqrt{x^2 + y^2}}{|x|} = e$$

or $(e^2 - 1)x^2 = y^2$ which is a pair of straight lines through the origin (focus).

1.5 RECOGNISATION OF CONICS

The equation of conics represented by the general equation of second degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \qquad ...(1)$$

can be recognised easily by the condition given in the tabular form. For this, first we have to find discriminant of the equation.

We know that the discriminant of above equation is represented by Δ where

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

Case I : When $\Delta = 0$

In this case equation (1) represents the **Degenerate conic** whose nature is given in the following table :

Condition	Nature of Conic
$\Delta = 0$ and $ab - h^2 = 0$	A pair of coincident lines
$\Delta = 0$ and $ab - h^2 \neq 0$	A pair of intersecting straight lines.
$\Delta = 0$ and $ab - h^2 < 0$	Real or imaginary pair of straight lines.
$\Delta = 0$ and $ab - h^2 > 0$	Point.

Case II : When $\Delta \neq 0$

In this case equation (1) represents the **Non-degenerate conic** whose nature is given in the following table :

Condition	Nature of Conic
$\Delta \neq 0$, $h = 0$, $a = b$	a circle
$\Delta \neq 0$, $ab - h^2 = 0$	a parabola
$\Delta \neq 0$, $ab - h^2 > 0$	an Ellipse or empty set
$\Delta \neq 0$, $ab - h^2 < 0$	a Hyperbola
$\Delta \neq 0$, $ab - h^2 < 0$ and $a + b = 0$	a Rectangular hyperbola

Illustration 1

If the equation $x^2 - y^2 - 2x + 2y + \lambda = 0$ represents a degenerate conic then find the value of λ.

Solution :

For degenerate conic $\Delta = 0$

Comparing the given equation of conic with

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$a = 1,\ b = -1,\ h = 0,\ g = -1,\ f = 1,\ c = \lambda$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

or $$(1)(-1)(\lambda) + 0 - 1 \quad (1)^2 + 1 \quad (-1)^2 - \lambda(0)^2 = 0$$

or $$-\lambda - 1 + 1 = 0 \quad \therefore \quad \lambda = 0$$

Illustration 2

If the equation $x^2 + y^2 - 2x - 2y + c = 0$ represents an empty set then find the value of c.

Solution :

For empty set $\Delta \neq 0$ and $ab - h^2 > 0$

Now comparing the given equation of conic with

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c' = 0$$

then $a = 1,\ h = 0,\ b = 1,\ g = -1,\ f = -1,\ c' = c$

. $$ab - h^2 > 0$$

$\therefore$ $$(1)(1) - 0 > 0$$

$$1 > 0 \qquad \text{which is true}$$

and $$\Delta = abc' + 2fgh - af^2 - bg^2 - c'h^2 \neq 0$$

or $$(1)(1)(c) + 0 - 1 \quad (-1)^2 - 1 \quad (-1)^2 - 0 \neq 0$$

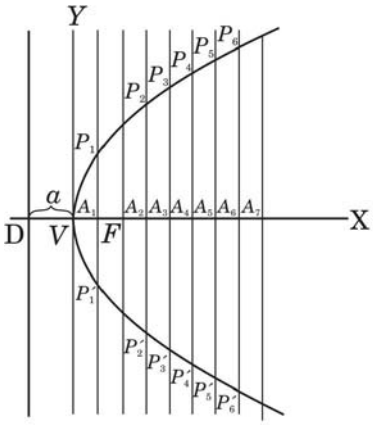
or $$c - 2 \neq 0$$

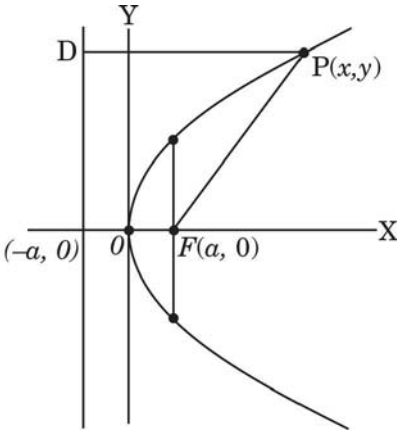
$\therefore$ $$c \neq 2$$

Hence $$c \in R \sim (2)$$

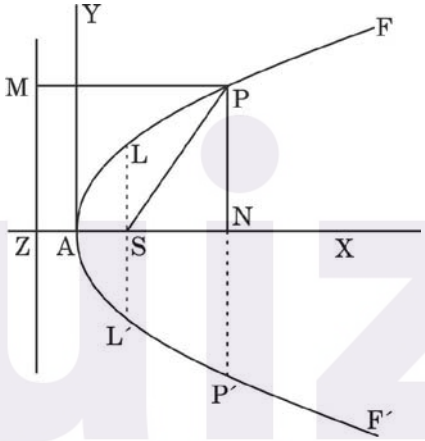
2.1 A geometric construction of the parabola

If several lines are drawn, all parallel to the directrix and on the same side of it as the focus, such that the distance from the directrix to each line is not less than a units, it is possible to construct many points of a parabola. In Fig., for example, the several points on the curve are located inthe followng manner: With $|A_1D|$ as a radius and F as a center, swing ares intersecting the perpendicular through A_1 at the points P_1 and P_1', respectively. Perform similar operations with $|A_2D|$, $|A_3D|$, etc., as radii and F as a center; and obtain successively the points P_2 and P_2, P_3 and P_3', etc..





Let S be the fixed point and ZM the directrix. We require therefore, the locus of a point P which moves so that its distance from S is always equal to PM, its perpendicular distance from ZM.



Draw SZ perpendicular to the directrix and bisect SZ in the point A; produce ZA to X.

The point A is clearly a point on the curve and is called the Vertex of the Parabola.

Take A as origin, AX as the axis of x, and AY, perpendicular to it, as the axis of y.

Let the distance ZA, or AS, be called a, and let P be any point on the curve whose coordinates are x and y.

Join SP, and draw PN and PM perpendicular respectively to the axis and directrix.

We have the, $SP^2 = PM^2$,

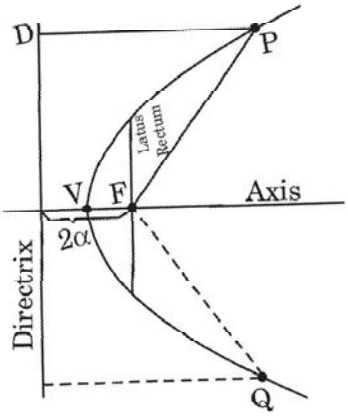
i.e., $$(x - a)^2 + y^2 = ZN^2 = (a + x)^2,$$

$\therefore$ $$\boxed{y^2 = 4ax} \quad ...(1)$$

This is also **standard form of parabola.**

Note :

The parabola is symmetric with respect to the line through the focus perpendicular to the directrix, since the image Q of every point P in the parabola with respect to this line is also a point on the parabola.



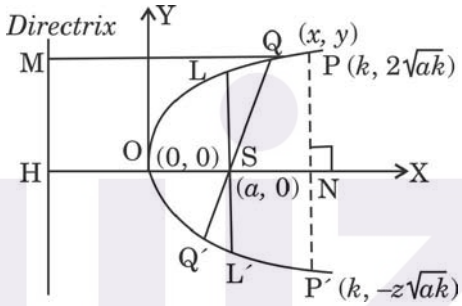
2.2 Some terms related to parabola :

(a) **Axis :** The axis of a parabola is the straight line through the focus and perpendicular to the directrix. In the parabola $y^2 = 4ax$, x-axis is the axis.

If the point (x, y) lies on the parabola $y^2 = 4ax$, the point (x, – y) also lies on it. Hence the parabola is symmetrical about its axis.

(b) **Vertex :** The point of intersection of the parabola and its axis is called its vertex.

In the parabola $y^2 = 4ax$, O(0, 0) i.e., the origin is the vertex.



(c) **Double ordinate :** If PN be drawn perpendicular tothe axis and produced to meet the curve again at P´, then PP´ is called a double ordinate.

(d) **Latus Rectum :** The double ordinate LSL´ which passes through the focus is called the latus rectum of the parabola.

The equation of the latus rectum of parabola $y^2 = 4ax$, is x = a.

Solving x = a and $y^2 = 4ax$, we find the co-ordinates of the extremities of the latus rectum which are L(a, 2a) and L´(a, – 2a).

Since LS = L´S = 2a, therefore length of the latus rectum LL´ = 4a.

(e) **Chord :** A cord of a parabola is the line segment joining any two points on the parabola.

(f) **Focal chord :** a chord of a parabola passing through the focus is called a focal chord. In the given figure QQ´ and LL´ are focal chords.

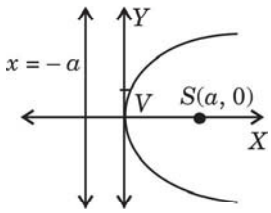
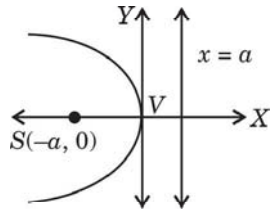
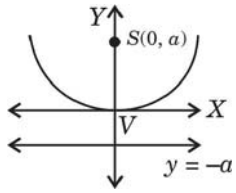
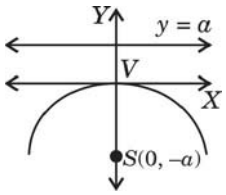
(g) **Focal distance :** The focal distance of any point Q(x, y) on the parabola is its distance from the focus i.e., QS.

Also QS = QM = distance of Q from the directrix. {as e = 1}

For the parabola $y^2 = 4ax$, a > 0.

(i) focus is (a, 0) (ii) Vertex is (0, 0) (iii) directrix is x = – a (iv) axis is x-axis i.e., y = 0 (v) length of latus rectum = 4a (vi) equation of latus rectum is x = a (vii) extremities of latus rectum are (a, 2a) and (a, – 2a) (viii) focal distance of Q(x, y) is (x + a).

There are other three forms also. All the four forms are listed as follows :

Equation	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Graph				
Vertex	(0, 0)	(0, 0)	(0, 0)	(0, 0)
Focus	(a, 0)	(– a, 0)	(0, a)	(0, – a)
Equation of directrix	x = – a	x = a	y = – a	y = a
Equation of the axis	y = 0	y = 0	x = 0	x = 0
Equation of tangent at the vertes	x = 0	x = 0	y = 0	y = 0
Parametric equations	$(at^2, 2at)$	$(-at^2, 2at)$	$(2at, at^2)$	$(2at, -at^2)$

If the vertex is not the origin and the axis and directrix are parallel to the coordinate axes, then the equation of parabola with vertex at (h, k) can be obtained by using translation of axes as follows :

Form	**Vertex**	**Focus**	**Equation of directrix**	**Equation of axis**	**Tangent at the vertex**
$(y - k)^2 = 4a(x - h)$	(h, k)	(h + a, k)	x = h – a	y = k	x = h
$(y - k)^2 = -4a(x - h)$	(h, k)	(h – a, k)	x = h + a	y = k	x = h
$(x - h)^2 = 4a(y - k)$	(h, k)	(h, k + a)	y = k – a	x = h	y = k
$(x - h)^2 = -4a(y - k)$	(h, k)	(h, k – a)	y = k + a	x = h	y = k

Note : in all the above cases a > 0.

Illustration 3

Find the equation of a parabola whose focus is (– 2, 3) and whose directrix is x + y + 5 = 0.

Solution :

Let P(h, k) be any point on the parabola then by definition the distance of P from (– 2, 3) must be equal to perpendicular distance of (h, k) from the line x + y + 5 = 0

$$\Rightarrow \quad \sqrt{(h+2)^2 + (k-3)^2} = \left|\frac{h+k+5}{\sqrt{2}}\right|$$

$\Rightarrow \qquad 2[(h + 2)^2 + (k - 3)^2] = (h + k + 5)^2$

On simplifying and replacing h, k by x, y we get the required equation of the parabola as

$$x^2 + y^2 - 2xy - 2x - 22y + 1 = 0$$

Illustration 4

Find the equation of the parabola whose vertex is (–1, 2) and focus is (3, 4).

Solution :

Since the vertex V(– 1, 2) lies on the parabola it must be equidistant from focus F(3, 4) and from directrix. If the line joining F and V (which is the axis of the parabola) meets the directrix at M(h, k) then V must be midpoint of FM.

$$\Rightarrow \qquad -1 = \frac{h+3}{2},\ 2 = \frac{k+4}{2} \Rightarrow \text{M is } (-5, 0)$$

Since directrix is perpendicular to the axis FM whose slope is $\frac{4-2}{3+1} = \frac{1}{2}$

$\Rightarrow$ Equation of directrix is $y - 0 = -2(x + 5)$ Or $2x + y + 10 = 0$

Since focus and directrix are known the equation of the parabola (as in the last example), is

$$\sqrt{(x-3)^2 + (y-4)^2} = \left(\frac{2x + y + 10)}{\sqrt{5}}\right)^2$$

which simplifies to $x^2 + 4y^2 - 4xy - 70x - 60y + 25 = 0$

Note :

We must note that in the equation of a parabola the second degree terms form a perfect square (i.e., $h^2 = ab$ in usual notations).

2.3 Reduction of equation of a parabola into standard form

- The equation $(y - \beta)^2 = 4a(x - \alpha)$ can be reduced to the standard form by the transformations $x - \alpha = X$, $y - \beta = Y$.

  The equation becomes $Y^2 = 4aX$, which is the standard form in X, Y coordinates.

  $(y - \beta)^2 = 4a(x - \alpha)$ is the form of equation of a parabola whose axis of symmetry is parallel to the x-axis.
- The equation $(ax + by + c)^2 = bx - ay + c'$ can be reduced to the standard form by the transoformations

$$\frac{ax + by + c}{\sqrt{a^2 + b^2}} = Y,\ \frac{bx - ay + c'}{\sqrt{b^2 + a^2}} = X$$

[**Note :** The lines $ax + by + c = 0$ and $bx - ay + c' = 0$ are perpendicular to each other.]

The equation becomes $Y^2 = \frac{1}{\sqrt{a^2 + b^2}}$ X which is the standard form in X, Y coordinates.

Illustration 5

Reduce the parabola $2y^2 + y + 5x + 1 = 0$ to the standard form and hence determine its axis, vertex, focus and equation of its directrix.

Solution :

We have $2y^2 + y = -5x - 1$

On dividing by 2 we get

$$y^2 + \frac{1}{2}y = -\frac{5}{2}x - \frac{1}{2} \Rightarrow y^2 + \frac{1}{2}y + \frac{1}{16} = -\frac{5}{2}x - \frac{1}{2} + \frac{1}{16}$$

$$\Rightarrow \left(y + \frac{1}{4}\right)^2 = -\frac{5}{2}x\frac{7}{16} \Rightarrow \left(y + \frac{1}{4}\right)^2 = -\frac{5}{2}\left(x + \frac{7}{40}\right)$$

On putting $X = x + \frac{7}{40}$, $Y = y + \frac{1}{4}$ (*)

We get $Y^2 = -\frac{5}{2}X$ (**)

$\Rightarrow 4a = -\frac{5}{2}$, $a = -\frac{5}{8}$ $\Rightarrow$ Length of latus rectum is $\frac{5}{8}$.

Now vertex of the parabola (**) is (0, 0).

$\Rightarrow$ Vertex of the original parabola is given by

$0 = x + \frac{7}{40}$, $0 = y + \frac{1}{4}$ i.e., the vertex is $\left(-\frac{7}{40}, \frac{-1}{4}\right)$

Again focus of the parabola (**) is (a, 0) i.e. $\left(-\frac{5}{8}, 0\right)$.

$\Rightarrow$ Focus of original parabola is given by $-\frac{5}{8} = x + \frac{7}{40}$, $0 = y + \frac{1}{4}$

$\Rightarrow$ Focus is $\left(-\frac{4}{5}, \frac{1}{4}\right)$

The axis and directrix of the original parabola are clearly given by $Y = 0$, $X + a = 0$

or in old system $y + \frac{1}{4} = 0$, $x + \frac{7}{40} - \frac{5}{8} = 0$ which reduce to, $4y + 1 = 0$ and $20x - 9 = 0$.

Trick : This can be easily noticed that the latus rectum of the parabola $Ay^2 + By + Cx + D = 0$ is $\left|\frac{C}{A}\right|$.

Illustration 6

Find the equation of the parabola whose vertex is at (2, 1) and the directrix is $x = y - 1$.

Solution :

The axis of the parabola is a line $\perp$ to the directrix and passing through the vertex. The equation of a line $\perp$ to $x - y + 1 = 0$ is

$$x + y + \lambda = 0$$

This will pass through (2, 1) if

$$2 + 1 + \lambda = 0 \Rightarrow \lambda = -3.$$

So the equation of the axis is

$$x + y - 3 = 0 \quad ...(i)$$

The equation of the directrix is

$$x - y + 1 = 0 \quad ...(ii)$$

Solving (i) and (ii),we get

$$x = 1, y = 2.$$

So, the coordinates of K are(1, 2).

Let (x_1, y_1) be the coordinates of the focus S. Then A is the mid-point of KS

$$\therefore \quad \frac{x_1+2}{2} = 2 \text{ and } \frac{y_1+2}{2} = 1$$

$$\Rightarrow \quad x_1 = 3 \text{ and } y_1 = 0$$

So, the coordinates of the focus S are (3, 0).

Let P(x, y) be a point on the parabola. Then

$$PS = PM$$

$$\Rightarrow \quad PS^2 = PM^2$$

$$\Rightarrow \quad (x-3)^2 + (y-0)^2 = \left(\frac{x-y+1}{\sqrt{1^2+(-1)^2}}\right)^2$$

$$\Rightarrow \quad 2(x^2 + y^2 - 6x + 9) = x^2 + y^2 + 1 - 2xy + 2x - 2y$$

$$\Rightarrow \quad x^2 + y^2 - 14x + 2y + 2xy + 17 = 0$$

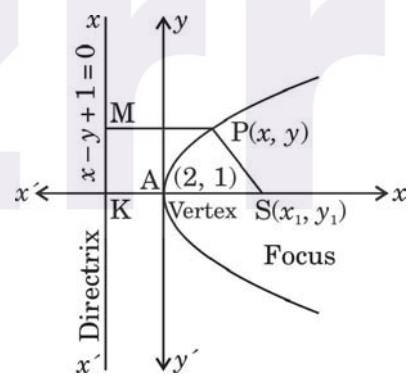


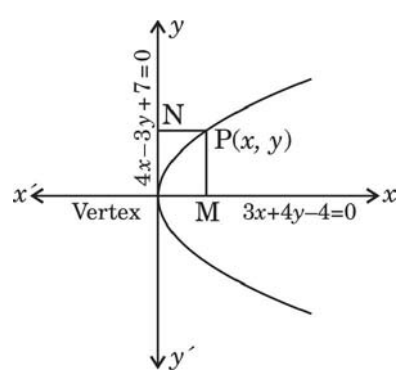
Illustration 7

Find the equation of the parabola whose latusrectum is 4 units, axis is the line $3x + 4y - 4 = 0$ and the tangent at the vertex is the line $4x - 3y + 7 = 0$.

Solution :

Let P (x, y) be any point on the parabola and let PM and PN be perpendiculars from P on the axis and tangent at the vertex respectively. Then

$$PM^2 = \textbf{(Latusrectum) (PN)}$$



$$\Rightarrow \quad \left(\frac{3x+4y-4}{\sqrt{3^2+4^2}}\right)^2 = 4\left(\frac{4x-3y+7}{\sqrt{4^2+(-3)^2}}\right)$$

$$\Rightarrow \quad (3x + 4y - 4)^2 = 20\ (4x - 3y \quad 7).$$

This is the equation of the required parabola.

Illustration 8

A double ordinate of the parabola $y^2 = 4ax$ is of length $8a$. Prove that the lines from the vertex to its ends are at right angles.

Solution :

Let PQ be the double ordinate of length 8a of the parabola $y^2 = 4ax$. Then PR = QR = 4a.

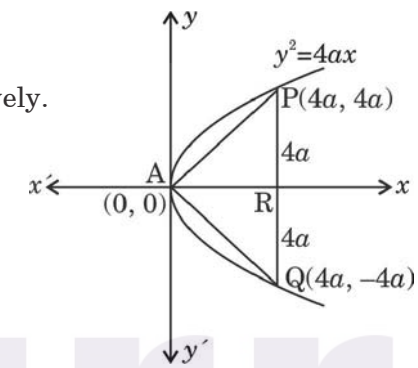
Let AR = x_1. Then the coordinates of P and Q are $(x_1, 4a)$ and $(x_1, -4a)$ respectively.

Since P lies on $y^2 = 4ax$.

$$\therefore \quad (4a)^2 = 4ax_1 \Rightarrow x_1 = 4a.$$

So, coordinates of P and Q are (4a, 4a) and (4a, – 4a) respectively.

Also, the coordinates of the vertex A are (0, 0).



$$\therefore \quad m_1 = \text{slope of AP} = \frac{4a-0}{4a-1} = 1,$$

$$\text{and,} \quad m_2 = \text{slope of AQ} = \frac{-4a-0}{4a-0} = -1$$

Clearly, $m_1m_2 = -1$. Hence, AP $\perp$ AQ.

Illustration 9

Prove that the equation to the parabola whose vertex and focus are on the x-axis at a distance a and a′ from the origin respectively is $y^2 = 4(a' - a)(x - a)$.

Solution :

Let O, A and S be the origin, vertex and focus of the parabola respectively. Then, OA = a, OS = a′. Therefore, the coordinates of S are (a′, 0). Let KK′ be the directrix of the required parabola.

Suppose SA produced meets the directrix at Z. Let the coordinates of Z be (x_1, y_1). Then.

$$\frac{x_1+a'}{2} = a \text{ and } \frac{y_1+0}{2} = 0 \qquad [\ .\ \text{A is the mid-point of SZ}]$$

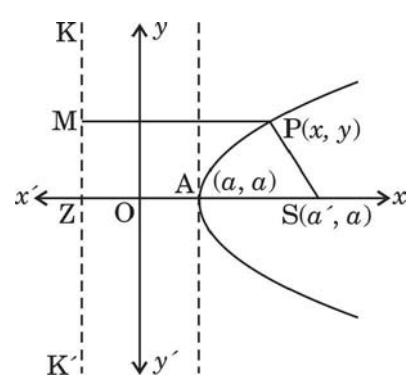
$\Rightarrow x_1 = 2a - a'$ and $y_1 = 09$

So, the equation of the directrix KK′ is $x = x_1$ i.e., $x = 2a - a'$.

Let P(x, y) be any point on the parabola. Then

SP = PM

$$\Rightarrow \sqrt{(x-a')^2+(y-0)^2} = \left|\frac{x-2a+a'}{\sqrt{1+0}}\right|$$



$\Rightarrow (x - a')^2 + y^2 = (x - 2a + a')^2$

$\Rightarrow (x - a')^2 + y^2 = [(x - a') - 2(a - a')]^2$

$\Rightarrow (x - a')^2 + y^2 = (x - a')^2 + 4(a - a')^2 - 4(x - a')(a - a')$

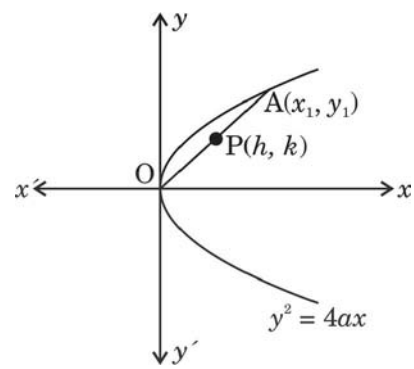
$\Rightarrow y^2 = 4(a - a')\{(a - a') - (x - a')\} \Rightarrow y^2 = 4(a' - a)(x - a)$.

Illustration 10

Find the locus of the middle points of all chords of the parabola $y^2 = 4ax$ which are drawn through the vertex.

Solution :

Let OA be a chord, drawn through the vertex and P(h, k) be its mid-point. Let the coordinates of A be (x_1, y_1). Then,



$\frac{x_1 + 0}{2} = h, \ \frac{y_1 + 0}{2} = k.$

$\Rightarrow x_1 = 2h$ and $y_1 = 2k$

So, the coordinates of A are (2h, 2k).

Since A lies on $y^2 = 4ax$.

$\therefore (2k)^2 = 4a(2h) \Rightarrow k^2 = 2ah$

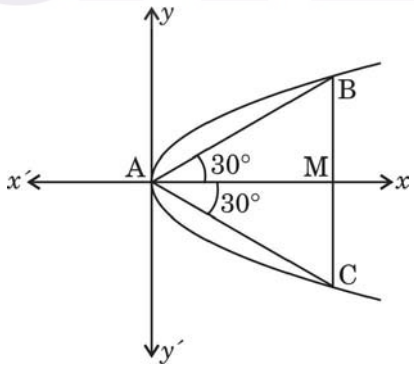
Hence, the locus of (h, k) is $y^2 = 2ax$.

Illustration 11

An equilateral triangle is inscribed in the parabola $y^2 = 4ax$ whose vertex is at the vertex of the parabola. Find the length of its side.

Solution :

Let AB = l. Then, AM = l cos 30 $= \frac{l\sqrt{3}}{2}$ and BM = l sin 30 $= \frac{l}{2}$



So, the coordinates of B are $\left(\frac{l\sqrt{3}}{2}, \frac{l}{2}\right)$

Since, B lies on $y^2 = 4ax$.

$\therefore \frac{l^2}{4} = 4a\left(\frac{l\sqrt{3}}{2}\right) \Rightarrow l = 8a\sqrt{3}$

Illustration 12

If y_1, y_2, y_3 be the ordinates of a vertices of the triangle inscribed in a parabola $y^2 = 4ax$, then show that the area of the triangle is $\frac{1}{8a}\ |(y_1 - y_2)\ (y_2 - y_3)\ (y_3 - y_1)|$.

Solution :

Let A (x_1, y_1), B(x_2, y_2) and C(x_3, y_3) be the vertices of ΔABC. Since (x_1, y_1), (x_2, y_2) and (x_3, y_3) lie on the parabola, therefore

$y_1^2 = 4ax_1$, $y_2^2 = 4ax_2$ and $y_3^2 = 4ax_3$

$\Rightarrow x_1 = \frac{y_1^2}{4a}$, $x_2 = \frac{y_2^2}{4a}$ and $x_3 = \frac{y_3^2}{4a}$

Now,

$$\text{Area of } \Delta ABC = \frac{1}{2}[x_1\ (y_2 - y_3) + x_2\ (y_3 - y_1) + x_3\ (y_1 - y_2)]$$

$$= \frac{1}{2}\left[\frac{y_1^2}{4a}(y_2 - y_3) + \frac{y_2^2}{4a}(y_3 - y_1) + \frac{y_3^2}{4a}(y_1 - y_2)\right]$$

$$= \frac{1}{8a}[y_1^2\ (y_2 - y_3) + (y_2^2 y_3 - y_2 y_3^2) - y_1\ (y_2^2 - y_3^2)]$$

$$= \frac{1}{8a}[y_1^2\ (y_2 - y_3) + y_2 y_3\ (y_2 - y_3) - y_1\ (y_2^2 - y_3^2)]$$

$$= \frac{1}{8a}(y_2 - y_3)\ [y_1^2 + y_2 y_3 - y_1\ (y_2 + y_3)]$$

$$= \frac{1}{8a}(y_2 - y_3)\ [(y_1^2 - y_1 y_2) + (y_2 y_3 - y_1 y_3)]$$

$$= \frac{1}{8a}(y_2 - y_3)\ [y_1\ (y_1 - y_2) - y_3\ (y_1 - y_2)]$$

$$= \frac{1}{8a}(y_2 - y_3)\ (y_1 - y_2)\ (y_1 - y_3)$$

$$= -\frac{1}{8a}(y_1 - y_2)\ (y_2 - y_3)\ (y_3 - y_1)$$

Hence,

$$\text{Area of } \Delta ABC = \frac{1}{8a}\ |(y_1 - y_2)\ (y_2 - y_3)\ (y_3 - y_1)|$$

Illustration 13

QQ´ is a double ordinate of a parabola $y^2 = 4ax$. Find the locus of its point of trisection.

Solution :

Let the double ordinate QQ´ meet the axis of the parabola.

$$y^2 = 4ax \qquad ...(1)$$

in N.

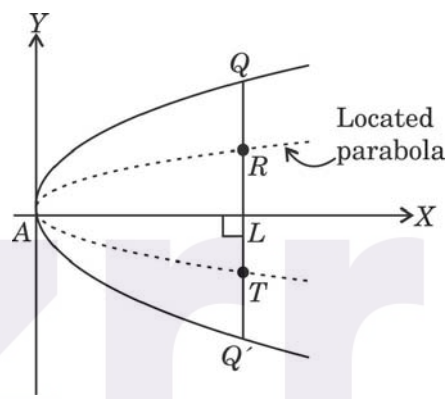
Let co-ordinate of Q be (x_1, y_1) then co-ordinate of Q´ be $(x_1, -y_1)$ since Q and Q´ lies on (1) then

$$y_1^2 = 4ax_1 \qquad ...(2)$$

Let R and T be the points of trisection of QQ´. Then the co-ordinates of R and T are

$$\left(\frac{1.x_1+2.x_1}{1+2}, \frac{1.(-y_1)+2.y_1}{1+2}\right) \text{ or } \left(x_1, \frac{y_1}{3}\right)$$

and $$\left(\frac{2.x_1+1.x_1}{2+1}, \frac{2.(-y_1)+1.y_1}{2+1}\right) \text{ or } \left(x_1, -\frac{y_1}{3}\right)$$



respectively.

Since R divide QQ´ in 1 : 2 (internally)

and T divide QQ´ in 2 : 1 (internally)

For locus, Let R (h, k) then

$x_1 = h$ and $\frac{y_1}{3} = k$ or $y_1 = 3k$

Substituting the values of x_1 and y_1 in (2), then

$$(3k)^2 = 4a\ (h)$$

or $$9k^2 = 4ah$$

$\therefore$ The required locus is $9y^2 = 4ax$ similarly, let T(h´, k´).

then $x_1 = h'$ and $-\frac{y_1}{3} = k'$ or $y_1 = -3k'$

Substituting the values of x_1 and y_1 in (2) then

$$(-3k')^2 = 4a(h')$$

or $$9k'^2 = 4ah'$$

$\therefore$ The required locus is $9y^2 = 4ax$

Hence the locus of point of trisection is

$$9y^2 = 4ax.$$

Alternative Method : Let R and T be the points of trisection of double ordinates QQ′. Let (h, k) be the co-ordinates of R.

then $AL = h$ and $RL = k$

$RT = RL + LT = k + k = 2k.$

Since $RQ = TR = Q'T = 2k$

$\therefore$ $LQ = LR + RQ = k + 2k = 3k$

thus the co-ordinates of Q are (h, 3k)

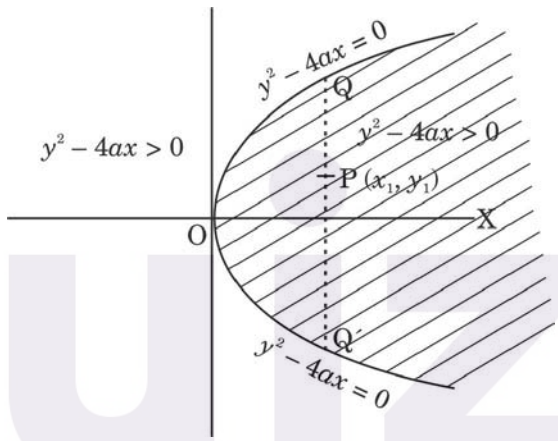
Since (h, 3k) lies on $y^2 = 4ax$

$\Rightarrow$ $9k^2 = 2ah$

Hence the locus of (h, k) is $9y^2 = 4ax$

2.4 The position of a point with respect to a parabola :

Let P (x_1, y_1) be a point in the plane if $y_1^2 - 4ax_1 < 0$ then the point P lies inside the parabola $y^2 = 4ax$. If $y_1^2 - 4ax_1 > 0$ then the point P lies outside the parabola.



The assertion can be easily proved. If $P(x_1, y_1)$ be any point withinthe parabola then produce the ordinate of P to a point $Q(x_1, y'_1)$ on the upper branch of the parabola. We must have

$y'_1 > |y_1| \Rightarrow y'^2_1 > y_1^2$

Now, as (x_1, y'_1) lies on the parabola $y^2 = 4ax$ we have $y'^2_1 = 4ax_1$

Thus $4ax_1 > y_1^2 \Rightarrow y_1^2 - 4ax_1 < 0$

If $P(x_1, y_1)$ lies in the left half plane then $x_1 < 0$ and $y_1^2 - 4ax_1 > 0$ is obvious. If P lies in the right half plane but does not lie within the parabola then $|y'_1| < |y_1|$.

which will reduce to $y_1^2 - 4ax_1 > 0$.

Note :

1. The point (x_1, y_1) lies inside, on or outside $y^2 = -4ax$ according as

$$y_1^2 + 4ax_1 <, = \text{ or } > 0$$

2. The point (x_1, y_1) lies inside, on or outside $x^2 = 4ay$ according as

$$x_1^2 - 4ay_1 <, =, \text{ or } > 0$$

3. The point (x_1, y_1) lies inside, on or outside $x^2 = -4ay$ according as

$$x_1^2 + 4ay_1 <, = \text{ or } > 0$$

Illustration 14

Show that the point (2, 3) lies outside the parabola $y^2 = 3x$.

Solution :

Let the point (h, k) = (2, 3)

We have $k^2 - 3h = 3^2 - 3.2$

$= 9 - 6 = 3 > 0$

$\therefore$ $k^2 - 3h > 0$

This shows that (2, 3) lies outside the parabola $y^2 = 3x$.

Illustration 15

Find the position of the point (–2, 3) with respect to the parabola $y^2 - 4y + 9x + 13 = 0$.

Solution :

Let the point (h, k) = (– 2, 2)

We have $k^2 - 4k + 9h + 13 = (2)^2 - 4(2) + 9(-2) + 13 = 4 - 8 - 18 + 13$

$= -9 < 0$

Hence $k^2 - 4k + 9h + 13 < 0$

Therefore the point (–2, 2) lies inside the parabola

$y^2 - 4y + 9x + 13 = 0$

3.1 PARAMETRIC EQUATION

The parametric form of the parabola $y^2 = 4ax$ is $(at^2, 2at)$. i.e. $x = at^2$, $y = 2at$.

You can see this by just putting the values in the equation.

3.2 EQUATION OF THE CHORD JOINING ANY TWO POINTS ON THE PARABOLA

Let $P(at_1^2, 2at_1)$, $Q(at_2^2, 2at_2)$ be any two points on the parabola $y^2 = 4ax$. Then, the equation of the chord joining these points is

$$y - 2at_1 = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2}(x - at_1^2)$$

or $$y - 2at_1 = \frac{2}{t_1 + t_2}(x - at_1^2)$$

or $$\boxed{y(t_1 + t_2) = 2x + 2at_1 t_2}$$

3.3 CONDITION FOR THE CHORD JOINING POINTS HAVING PARAMETERS t_1 AND t_2 TO BE A FOCAL CHORD

If the chord joining points $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ on the parabola passes through its focus, then (a, 0) satisfies the equation

$$y(t_1 + t_2) = 2x + 2at_1t_2$$

$\Rightarrow$ $$0 = 2a + 2at_1t_2$$

$\Rightarrow$ $$\boxed{t_1 t_2 = -1}$$

3.4 LENGTH OF THE FOCAL CHORD

Let P $(at_1^2, 2at_1)$ and Q $(at_2^2, 2at_2)$ be the end point of a focal chord. Then, $t_1 t_2 = -1$.

Now
$$PQ = \sqrt{(at_2^2 - at_1^2)^2 + (2at_2 - 2at_1)^2}$$
$$= a(t_2 - t_1)\sqrt{(t_2 + t_1)^2 + 4}$$
$$= a(t_2 - t_1)\sqrt{(t_2 + t_1)^2 - 4t_1t_2} \qquad [\because t_1 t_2 = -1]$$
$$= a(t_2 - t_1)\sqrt{(t_2 - t_1)^2}$$
$$= a(t_2 - t_1)^2$$

Thus, the length of a focal chord having parameters t_1 and t_2 for its end points is $a(t_2 - t_1)^2$.

Illustration 16

Show that the focal chord of parabola $y^2 = 4ax$ makes an angle α with the x-axis is of length $4a \operatorname{cosec}^2 \alpha$.

Solution :

Let P $(at_1^2, 2at_1)$ and Q$(at_2^2, 2at_2)$ be the end points of a focal chord PQ which makes an angle α with the axis of the parabola. Then,

$$PQ = a(t_2 - t_1)^2 \text{ and } \tan\alpha = \text{slope of } PQ = \frac{2}{t_1 + t_2}$$

Now, $PQ = a(t_2 - t_1)^2$

$\therefore$
$$PQ = a[(t_2 + t_1)^2 - 4t_1t_2]$$
$$= a[4\cot^2\alpha + 4] \quad [\because t_2 + t_1 = 2\cot\alpha \text{ and } t_1 t_2 = -1]$$
$$= 4a\operatorname{cosec}^2\alpha$$

Illustration 17

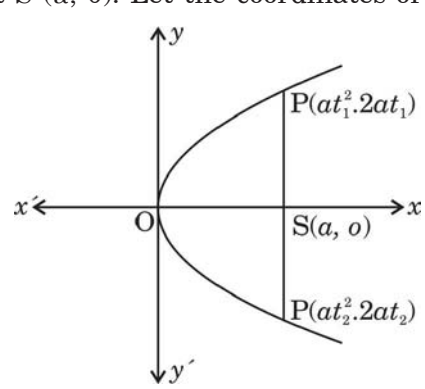
Show that the semi-latusrectum of the parabola $y^2 = 4ax$ is the harmonic mean between the segments of any focal chord of the parabola.

Solution :

Let PQ be a focal chord of the parabola $y^2 = 4ax$ having focus at S (a, 0). Let the coordinates of P and Q be $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$. Then, $t_1t_2 = -1$.

Now, SP = Focal distance of P
$$= a + at_1^2 = a(1 + t_1^2)$$

Similarly, $SQ = a(t_2^2 + 1)$
$$= a\left(\frac{1}{t_1^2} + 1\right) = a\frac{(1 + t_1^2)}{t_1^2} \qquad [\because t_1 t_2 = -1]$$



$$\therefore \left(\frac{1}{SP}+\frac{1}{SQ}\right)=\frac{1}{a(t_1^2+1)}+\frac{t_1^2}{a(t_1^2+1)}=\frac{1}{a} \Rightarrow \frac{SP+SQ}{SP.SQ}=\frac{1}{a}$$

$$\Rightarrow \frac{SP.SQ}{SP+SQ}=a \Rightarrow \frac{2\ SP.SQ}{SP+SQ}=2a$$

$\Rightarrow$ Harmonic mean of SP and SQ is the semi-latusrectum of the parabola $y^2 = 4ax$.

Trick : It follows from this example, that if l_1 and l_2 are the length segments of a focal chord of a parabola, then its latus rectum is $\frac{4l_1l_2}{l_1+l_2}$

4.1 INTERSECTION OF A LINE AND PARABOLA

For a parabola $y^2 = 4ax$ and a line $y = mx + c$. there are 3 cases possible :

1. The line and the parabola do not intersect.
2. The line and the parabola intersect at one point only. This is the case of touching the parabola i.e. tangent to parabola.
3. The line and parabola intersect at 2 points i.e., the line has a line segment which acts as chord to parabola.

The above 3 conditions are alzebraically equivalent to saying that the quadratic equation formed by the intersection of two curves will either have :

1. No roots.
2. Only one real root.
3. 2 distinct real roots.

So putting value of y, i.e. $y = mx + c$ in $y^2 = 4ax$

$$(mx + c)^2 = 4ax$$

$$\Rightarrow \quad m^2x^2 + c^2 + 2mcx = 4ax$$

$$\Rightarrow \quad m^2x^2 + (2mc - 4a)\,x + c^2 = 0 \qquad ...(i)$$

Discriminant of this equation

$$D^2 = (2mc - 4a)^2 - 4m^2c^2$$
$$= 4\ \{(mc - 2a)^2 - m^2c^2\}$$
$$= 16\ (a^2 - amc)$$

Now for our 3 cases

1. No real roots, $D < 0 \Rightarrow a < mc$ i.e. no intersection.
2. Only one real root, $D = 0 \Rightarrow a = mc$ i.e., tangent.
3. 2 real distinct roots $D > 0 \Rightarrow a > mc$ i.e., chord to parabola.

Let us discuss the case 3 first and find out the length of chord formed.

Case 3 : Let the line meet parabola in 2 distinct points (x_1, y_1) and (x_2, y_2). Then the length of chord will be :

$$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2} = \sqrt{(x_2-x_1)^2+(mx_2+c-mx_1-c)^2}$$

$$= \sqrt{(x_2 - x_1)^2 + m^2(x_2 - x_1)^2}$$

$$= \sqrt{1+m^2}\ \sqrt{(x_2 - x_1)^2}$$

$$= \sqrt{1+m^2}\ \sqrt{(x_1 + x_2)^2 - 4x_1x_2}$$

$$= \sqrt{1+m^2}\ \sqrt{\frac{4(mc-2a)^2}{m^4} - \frac{4c^2}{m^2}}$$

because from (i) $x_1 + x_2 = \dfrac{-2(mc-2a)}{m^2}$

$$x_1x_2 = \frac{c^2}{m^2}$$

$$= \frac{2}{m^2}\sqrt{1+m^2}\ \sqrt{(mc-2a)^2 - c^2m^2}$$

$$= \frac{2}{m^2}\sqrt{1+m^2}\ \sqrt{4a^2 - 4amc}$$

$$= \frac{4}{m^2}\sqrt{1+m^2}\ \sqrt{a^2 - amc}$$

$$= \frac{4}{m^2}\sqrt{1+m^2}\ \sqrt{a(a-mc)}$$

So the length of chord cut by the line $= \dfrac{4}{m^2}\sqrt{1+m^2}\ \sqrt{a(a-mc)}$

Now let us discuss the case 2.

Case 2 : Condition for tangency :

as done earlier, for this D = 0 then the condition is $\boxed{\text{'c'} = \dfrac{a}{m}}$

So putting this value of c in line to get the tangent equation

$$y = mx + \frac{a}{m}$$

Point of contact :

Putting $c = \dfrac{a}{m}$ in quadratic equation for finding the points on parabola.

$$m^2x^2 + \left(2m\left(\frac{a}{m}\right) - 4a\right)x + \frac{a^2}{m^2} = 0$$

$$m^2x^2 - 2ax + \frac{a^2}{m^2} = 0$$

$$\left(mx - \frac{a}{m}\right)^2 = 0$$

$$\Rightarrow \qquad x = \frac{a}{m^2}$$

Putting this equation of line to get value of y

$$y = mx + \frac{a}{m} = m\left(\frac{a}{m^2}\right) + \frac{a}{m} = \frac{2a}{m}$$

$\therefore$ Point of contact becomes : $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

Illustration 18

Prove that for any non-zero real m**, the line** $y = mx + \frac{3}{m}$ **is a real chord to the parabola** $y^2 = 16x$**. Find the value of** m **if length of the chord is 3.**

Solution :

In the given question $a = 4$, $c = \frac{3}{m}$, here $a > mc$ for any m, so line is a real chord.

Now length of the chord is $\frac{4}{m^2}\sqrt{1+m^2}\sqrt{a(a-mc)} = \frac{4}{m^2}\sqrt{1+m^2}\sqrt{4(4-3)}$

$$= \frac{8}{m^2}\sqrt{1+m^2} = 3 \text{ (given)}$$

$$\Rightarrow \qquad 64(1 + m^2) = 9m^4$$

$$\Rightarrow \qquad 9m^4 - 64m^2 - 64 = 0$$

$$\Rightarrow \qquad m^2 = 8 \Rightarrow m = \pm 2\sqrt{2}.$$

Illustration 19

For what value of k **does the line** $x + y = 1$ **touches the parabola** $y^2 = kx$ **?**

Solution :

We have $y = -x + 1$

If this line touches the parabola $y^2 = kx$, then

$$1 = \frac{k/4}{-1} \qquad \text{[Using } c = a/m\text{]}$$

$$\Rightarrow \qquad k = -4.$$

Illustration 20

Prove that the line $lx + my + n = 0$ will touch the parabola $y^2 = 4ax$ if $ln = am^2$.

Solution :

We hae $lx + my + n = 0 \Rightarrow y = \left(-\frac{1}{m}\right)x + \left(-\frac{n}{m}\right)$

This will touch $y^2 = 4ax$, if

$$\frac{-m}{n} = \frac{a}{(-l/m)} \qquad \text{[Using } c = a/m\text{]}$$

$$\Rightarrow \qquad nl = am^2$$

Illustration 21

Find the condition that the line $x \cos \alpha + y \sin \alpha = p$ touches the parabola $y^2 = 4ax$.

Solution :

We have $x \cos \alpha + y \sin \alpha = p$

$\Rightarrow y \sin \alpha = -x \cos \alpha + p \Rightarrow y = x(-\cot \alpha) + p \operatorname{cosec} \alpha$.

This will touch the parabola $y^2 = 4ax$, if

$$p \operatorname{cosec} \alpha = \frac{a}{-\cot \alpha} \qquad \text{[Using } c = a/m\text{]}$$

$\Rightarrow -p \cos \alpha = a \sin^2 \alpha \Rightarrow p \cos \alpha + a \sin^2 \alpha = 0$.

Illustration 22

Show that $y = mx + c$ touches the parabola $y^2 = 4a(x + a)$, if $c = am + \frac{a}{m}$.

Solution :

We have : $y^2 = 4a(x + a)$ and $y = mx + c$. Shifting the origin at $(-a, 0)$, we have :

$$x = X + (-a) \text{ and } y = Y + 0.$$

Substituting the values of x and y in the equations of the parabola and the line, we get

$$Y^2 = 4a\,X \text{ and } y = mX + c - am$$

The line will touch the parabola, if

$$c - am = \frac{a}{m} \qquad \text{[Using } c = am\text{]}$$

$$\Rightarrow \qquad c = am + \frac{a}{m}$$

5.1 EQUATION OF TANGENT IN VARIOUS FORMS

Type 1 : <u>POINT FORM</u> :

The equation of tangent to parabola at point $(x_1\ y_1)$ is given by $\mathbf{yy_1 = 2a\ (x + x_1)}$

Logic : for parabola $y^2 = 4ax$

Slope at point $(x_1\ y_1)$ is given by

$$2y\left(\frac{dy}{dx}\right) = 4a \Rightarrow \frac{dy}{dx} = \frac{4a}{2y} = \frac{2a}{y}$$

and $$\left.\frac{dy}{dx}\right]_{x_1 y_1} = \frac{2a}{y_1}$$

$\therefore$ Slope of tangent at $(x_1\ y_1) = \dfrac{2a}{y_1}$

$\therefore$ equation tangent becomes

$$y - y_1 = \frac{2a}{y_1}\ (x - x_1)$$

Solving this we get the required equation.

TRICK : This form can also be obtained by replacing x^2 y xx_1, y^2 by yy_1, x by $\dfrac{x+x_1}{2}$, y by $\dfrac{y+y_1}{2}$ and xy by $\dfrac{xy_1+x_1y}{2}$ in general equation of parabola.

Type 2 : <u>PARAMETRIC FORM</u> :

The equation of tangent in parametric form given the parametric point $(at^2, 2at)$ on parabola $y^2 = 4ax$ is

$$\boxed{ty = x + at^2}$$

just replace x by at^2 and y 2at in point form to get the answer

Type 3 : <u>SLOPE FORM</u> :

This one we have already when we discuss the condition for tangency i.e.

$$\boxed{y = mx + \frac{a}{m}}$$

Illustration 23

Find m for which the line $3y + x - m = 0$ is a tangent to the parabola $y^2 = -7x$.

Solution :

Putting the line in the form y = mx + c we get $y = -\dfrac{1}{3}x + \dfrac{m}{3}$

$\Rightarrow$ m and c are $-\dfrac{1}{3}$ and $\dfrac{m}{3}$ respectively.

Also, $4a = -7 \Rightarrow a = -\frac{7}{4}$

Applying $c = \frac{a}{m}$ we get $\frac{m}{3} = \frac{-7/4}{-1/3} \Rightarrow m = \frac{63}{4}$

Illustration 24

Find k for which the line $x + 2y + k = 0$ touches the parabola $y^2 + 4y + 4x = 0$.

Solution :

First Method :

The parabola can be written as $y^2 + 4y = 4x$

$\Rightarrow \quad (y + 2)^2 = -4(x - 1)$

Put $X = x - 1$, $y + 2 = Y$ we get

$$Y^2 = -4X \Rightarrow a = -1$$

Also line, in new coordinate system, is

$$X + 1 + 2(Y - 2) + k = 0 \Rightarrow Y = -\frac{1}{2}X + \frac{3-k}{3}$$

For tangency $c = \frac{9}{m} \Rightarrow \frac{3-k}{2} = \frac{-1}{-1/2} \Rightarrow k = -1$

Second Method :

Eliminating x between the two equations we get

$y^2 + 4y + 4(-2y - k) = 0$

$\Rightarrow \quad y^2 - 4y - 4k = 0$

For tangency this equation must have equal roots

$\Rightarrow \quad 16 + 16k = 0 \Rightarrow k = -1$

NOTE :

The second method is always applicable. It is even applicable in the case when the equation of the parabola contains xy term.

Illustration 25

Find the common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$.

Solution :

Any tangent to the parabola $y^2 = 8x$ may be taken as

$y = mx + \frac{a}{m}$ OR $y = mx + \frac{2}{m}$

$\Rightarrow m^2x - my + 2 = 0$ (*)

If this a tangent to the circle $x^2 + y^2 = 2$, the perpendicular length from centre (0, 0) of the circle on the line (*) must be equal to radius $\sqrt{2}$ of the circle.

$$\Rightarrow \left|\frac{2}{\sqrt{m^4+m^2}}\right| = \sqrt{2} \Rightarrow m^4 + m^2 - 2 = 0$$

$\Rightarrow (m^2 + 2)(m^2 - 1) = 0 \Rightarrow m = \pm 1$ ($m^2 + 2 = 0$ is not possible)

$\Rightarrow$ Common tangents to the circle and the parabola are $y = x + 2$ and $y = -x - 2$

5.2 POINT OF INTERSECTION OF TANGENTS

Point of intersection of tangents at the points 't_1' and 't_2' is $(at_1t_2, a(t_1 + t_2))$.

NOTE :

1. The Geometric mean of the x co-ordinates of P and Q (i.e., $\sqrt{at_1^2 \times at_2^2} = at_1t_2$) is the x co-ordinate of the point of intersection of tangents at P and Q on the parabola.
2. The Arithmetic mean of the y-co-ordinates of P and Q $\left(\text{i.e.,} \frac{2at_1 + 2at_2}{2} = a(t_1 + t_2)\right)$ is the y-co-ordinate of the point of intersection of tangents at P and Q on the parabola.

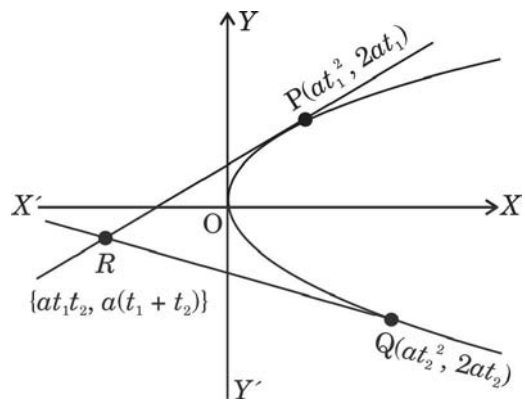


Illustration 26

Show that the tangents at the extremities of any focal chord of a parabola intersect at right angles at the directrix.

Solution :

The equations of tangents at two points $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ are

$$t_1y = x + at_1^2 \quad \text{...(i)}$$

and,

$$t_2y = x + at_2^2 \quad \text{...(ii)}$$

The coordinates of the point of intersection of (i) and (ii) are

$$(at_1 t_2, a(t_1 + t_2))$$

Since PQ is a focal chord. Therefoere, $t_1 t_2 = -1$.

Now, slope of (i) slope of (ii) $= \frac{1}{t_1} \times \frac{1}{t_2} = \frac{1}{t_1t_2} = \frac{1}{-1} = -1$

Hence, tangents at P and Q are at right angle.

The coordinates of the point of intersection are $(-a, a(t_1 + t_2))$. Clearly, this point lies on the directrix.

Illustration 27

Show that the locus of the point of intersection of perpendicular tangents to the parabola is its directrix.

Solution :

Let P(h, k) be the point of intersection of tangents at points $Q(at_1^2, 2at_1)$ and $R(at_2^2, 2at_2)$ on the parabola $y^2 = 4ax$.

Then, $h = at_1t_2$ and $k = a(t_1 + t_2)$.

The equations of tangents at Q and R are

$t_1y = x + at_1^2$ and $t_2y = x + at_2^2$ respectively.

These two will be perpendicular, if

$$\frac{1}{t_1} \times \frac{1}{t_2} = -1 \Rightarrow t_1 t_2 = -1$$

Putting $t_1 t_2 = -1$ in $h = at_1 t_2$, we get $h = -a$.

So, the locus of (h, k) is $x = -a$, which is the directrix of the parabola.

Illustration 28

Prove that the area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points. [I.I.T. 1996]

Solution :

Let $A(at_1^2, 2at_1)$, $B(at_2^2, 2at_2)$ and $C(at_3^2, 2at_3)$ be three points on the parabola $y^2 = 4ax$. Then,

$$\Delta_1 = \text{Area of } \Delta ABC$$

$$= \frac{1}{2}\begin{vmatrix} at_1^2 & 2at_1 & 1 \\ at_2^2 & 2at_2 & 1 \\ at_3^2 & 2at_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}(2a^2)\begin{vmatrix} t_1^2 & t_1 & 1 \\ t_2^2 & t_2 & 1 \\ t_3^2 & t_3 & 1 \end{vmatrix}$$

$$= a^2 \begin{vmatrix} t_1^2 & t_1 & 1 \\ t_2^2 - t_1^2 & t_2 - t_1 & 0 \\ t_3^2 - t_1^2 & t_3 - t_1 & 0 \end{vmatrix} \qquad \left[\begin{array}{c}\text{Applying } R_2 \to R_2 - R_1 \\ \text{and } R_3 \to R_3 - R_1\end{array}\right]$$

$$= a^2 (t_2 - t_1 (t_3 - t_1) \begin{vmatrix} t_1^2 & t_1 & 1 \\ t_2 + t_1 & 1 & 0 \\ t_3 + t_1 & 1 & 0 \end{vmatrix}$$

$$= a^2 (t_2 - t_1)(t_3 - t_1) \begin{vmatrix} t_2 + t_1 & 1 \\ t_3 + t_1 & 1 \end{vmatrix}$$

$$= a^2 (t_2 - t_1)(t_3 - t_1)(t_2 - t_3)$$

$$= - a^2 (t_1 - t_2)(t_2 - t_3)(t_3 - t_1)$$

$$= a^2 (t_1 - t_2)(t_2 - t_3)(t_3 - t_1) \text{ in magnitude.}$$

Suppose the tangents at A, B, C taken in pairs intersect at P, Q and R. Then, coordinates of P, Q and R are $(at_1t_2, a(t_1 + t_2))$, $(at_2t_3, a(t_2 + t_3))$ and $(at_3t_1, a(t_3 + t_1))$ respectively.

$\therefore \quad \Delta_2$ = Area of ΔPQR

$$= \frac{1}{2} \begin{vmatrix} at_1t_2 & a(t_1+t_2) & 1 \\ at_2t_3 & a(t_2+t_3) & 1 \\ at_3t_1 & a(t_3+t_1) & 1 \end{vmatrix}$$

$$= \frac{1}{2} a^2 \begin{vmatrix} t_1t_2 & t_1+t_2 & 1 \\ t_2t_3 & t_2+t_3 & 1 \\ t_3t_1 & t_3+t_1 & 1 \end{vmatrix}$$

$$= \frac{1}{2} a^2 \begin{vmatrix} t_1t_2 & t_1+t_2 & 1 \\ t_2(t_3-t_1) & t_3-t_1 & 0 \\ t_1(t_3-t_2) & t_3-t_2 & 0 \end{vmatrix} \quad \begin{bmatrix} \text{Applying } R_2 \to R_2 - R_1 \\ \text{and } R_3 \to R_3 - R_1 \end{bmatrix}$$

$$= \frac{1}{2} a^2 (t_3-t_1)(t_3-t_2) \begin{vmatrix} t_1t_2 & t_1+t_2 & 1 \\ t_2 & 1 & 0 \\ t_3 & 1 & 0 \end{vmatrix}$$

$$= \frac{1}{2} a^2 (t_3 - t_1)(t_3 - t_2)(t_2 - t_1) = \frac{1}{2} a^2 (t_1 - t_2)(t_2 - t_3)(t_3 - t_1)$$

Clearly, $\Delta_1 = 2\Delta_2$.

Illustration 29

If the tangents at the points P and Q on a parabola meet in T, prove that $ST^2 = SP \cdot SQ$ i.e. **ST is the geometric mean of the focal distances of P and Q.**

Solution :

Let P $(at_1^2, 2at_1)$ and Q $(at_2^2, 2at_2)$ be two points on the parabola $y^2 = 4ax$. The tangents at P and Q intersect at $T(at_1 t_2, a(t_1 + t_2))$.

$\therefore \quad ST^2 = (a\, t_1t_2 - a)^2 + \{a(t_1 + t_2) - 0)\}^2$

$$= a^2 [(t_1t_2 - 1)^2 + (t_1 + t_2)^2] = a^2 [t_1^2 + t_2^2 + t_1^2 t_2^2 + 1]$$

$$= a^2 (t_1^2 + 1)(t_2^2 + 1) = a(t_1^2 + 1) \cdot a(t_2^2 + 1)$$

$$= (at_1^2 + a)\,(at_2^2 + 1)$$

$$= SP \,.\, SQ \qquad \left[\begin{array}{l} \because\ SP = x + a = at_1^2 + a \\ \text{and, } SQ = x + a = at_2^2 + a \end{array}\right]$$

Illustration 30

Show that the area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

Solution :

Let the three points on the parabola be $(at_1^2, 2at_1)$, $(at_2^2, 2at_2)$ and $(at_3^2, 2at_3)$.

The area of the triangle formed by these points

$$= \frac{1}{2}\begin{vmatrix} at_1^2 & 2at_1 & 1 \\ at_2^2 & 2at_2 & 1 \\ at_3^2 & 2at_3 & 1 \end{vmatrix} = -\,a^2\,(t_2 - t_3)\,(t_3 - t_1)\,(t_1 - t_2)$$

The intersections of the tangents at these points are the points

$\{at_2t_3,\ a(t_2 + t_3)\}$, $\{at_3t_1,\ a(t_3 + t_1)\}$, and $\{at_1t_2,\ a(t_1 + t_2)\}$

The area of the triangle formed by these three points $= \frac{1}{2}\begin{vmatrix} at_2t_3 & a(t_2+t_3) & 1 \\ at_3t_1 & a(t_3+t_1) & 1 \\ at_1t_2 & a(t_1+t_2) & 1 \end{vmatrix}$

$$= \frac{1}{2}\{at_2t_3(at_3 - at_2) + at_3t_1\,(at_1 - at_3) + at_1t_2\,(at_2 - at_1)\}$$

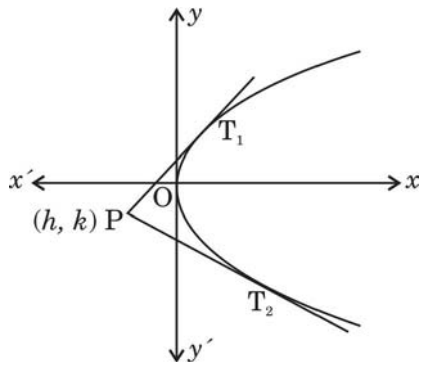
$$= \frac{1}{2}a^2\,(t_2 - t_3)\,(t_3 - t_1)\,(t_1 - t_2)$$

Since the absolute value of the first expression is double the second expression. The result follows.

5.3 NUMBER OF TANGENTS DRAWN FROM A POINT TO A PARABOLA

THEOREM : Two tangents can be drawn from a point to a parabola. The two tangents are real and distinct or coincident or imaginary according as the given point lies outside, on or inside the parabola.

PROOF : Let the parabola be $y^2 = 4ax$ and $P(h, k)$ be a point.



The equation of any tangent to the given parabola is

$$y = mx + \frac{a}{m}$$

If it passes through (h, k) then

$$k = mh + \frac{a}{m}$$

$$\Rightarrow \qquad m^2h - mk + a = 0$$

This equation, being a quadratic in m, gives two values of m. Corresponding to each of these values of m there is a tangent to the parabola.

Thus, two tangents can be drawn from a point to a parabola.

The tangents are real and distinct, coincident or imaginary according as the roots of quadratic in m are real and distinct, equal or imaginary i.e. according as

$$k^2 - 4ah > \text{ or } = \text{ or } < 0$$

i.e. (h, k) lies outside or on or side the parabola $y^2 = 4ax$.

Illustration 31

Prove that the locus of the point of intersection of tangents to the parabola $y^2 = 4ax$ which meet at an angle α is $(x + a)^2 \tan^2 \alpha = y^2 - 4ax$.

Solution :

Let P (h, k) be the point of intersection of tangents to the parabola $y^2 = 4ax$.

The equation of any tangent to $y^2 = 4ax$ is

$$y = mh + \frac{a}{m}$$

If it passes through (h, k), then

$$k = mh + \frac{a}{m} \Rightarrow m^2h - mk + a = 0$$

Let m_1, m_2 be the roots of this equation. Then,

$$m_1 + m_2 = \frac{k}{h} \text{ and } m_1 m_2 = \frac{a}{h}$$

Clearly, m_1, m_2 are the slopes of the tangents drawn from P.

$$\therefore \qquad \tan\alpha = \frac{m_1 - m_2}{1 + m_1 m_2} \Rightarrow \tan\alpha = \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2}$$

$$\Rightarrow \qquad \tan\alpha = \frac{\sqrt{k^2 - 4ah}}{a + h} \Rightarrow k^2 - 4ah = (a + h)^2 \tan^2\alpha$$

Hence, the locus of (h, k) is $y^2 - 4ax = (a + x)^2 \tan^2\alpha$

Illustration 32

Find the locus of the point of intersection of perpendicular tangents to the parabola $y^2 = 4ax$.

Solution :

Forming the equation as formed in the previous illustration i.e., $m^2h - mk + a = 0$

$$\Rightarrow \qquad m_1 m_2 = \frac{a}{h}$$

If the tangents are perpendicular, then

$$m_1 m_2 = -1 \Rightarrow \frac{a}{h} = -1 \Rightarrow h = -a$$

Hence, the locus of (h, k) is $x = -a$, which is the directrix of the parabola.

Illustration 33

Find the locus of the point of intersection of two tangents to the parabola $y^2 = 4ax$ such that they include at angle of 45 .

Solution :

From illustration 32, we have

$$y^2 - 4ax = (a + x)^2 \tan^2\alpha$$

Putting $\alpha = 45$, we get

$$y^2 - 4ax = (a + x)^2$$

Hence, the required locus is $y^2 - 4ax = (a + x)^2$

5.4 DIRECTOR CIRCLE :

The locus of the point of intersection of perpendicular tangents to a conic is known as its director circle.

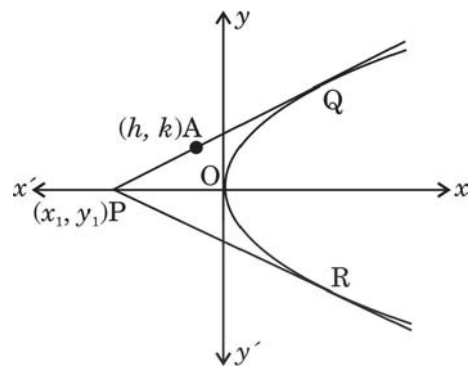
The director circle of a parabola is its directrix.

5.5 EQUATION OF THE PAIR OF TANGENTS FROM A POINT TO A PARABOLA

The combined equation of the pair of tangents drawn from a point to a parabola is $\boxed{SS' = \mathbf{T^2}.}$

PROOF : Let PQ and PR be the tangents drawn from the point $P(x_1, y_1)$ to the parabola $y^2 = 4ax$. Let A (h, k) be any point on either tangent through P. Then, equation of PA is

$$y - y_1 = \frac{k - y_1}{h - x_1}(x - x_1)$$



or, $$y = \left(\frac{k - y_1}{h - x_1}\right)x + \left(\frac{hy_1 - kx_1}{h - x_1}\right) \quad ...(i)$$

This equation should be of the form

$$y = mx + \frac{a}{m} \quad ...(ii)$$

Comparing (i) and (ii), we get

$$m = \frac{k - y_1}{h - x_1} \text{ and } \frac{a}{m} = \frac{hy_1 - kx_1}{h - x_1}$$

Eliminating m from these two, we obtain

$$a = \frac{k - y_1}{h - x_1} \times \frac{hy_1 - kx_1}{h - x_1} \Rightarrow a\,(h - x_1)^2 = (k - y_1)\,(hy_1 - kx_1)$$

Hence, the locus of the (h, k) i.e. equation of the pair of tangents, is

$$a(x - x_1)^2 = (y - y_1)\,(xy_1 - yx_1)$$

or, $$(y^2 - 4ax)\,(y_1^2 - 4ax) = (yy_1 - 2a\,(x + x_1)\}^2$$

or, $$SS' = T^2,$$

where $S = y^2 - 4ax$, $S' = y_1^2 - 4ax_1$ and $T = yy_1 - 2a\,(x + x_1)$.

REMARK : S = 0 is the equation of the curve, S´ is obtained from S by replacing x by x_1 and y by y_1 and T = 0 is the equation of the tangent.

Illustration 34

Find the angle between the tangents of the parabola $y^2 = 8x$, which are drawn from the point (2, 5).

Solution :

Let $y = mx + \frac{2}{m}$ be a tangent of parabola $y^2 = 8x$, if it passes through (2, 5),

then $5 = 2m + \frac{2}{m}$ or $2m^2 - 5m + 2 = 0$

roots of this quadratic equation in m will give slope of tangents of the parabola $y^2 = 8x$, which are passing through point (2, 5).

Angle between these tangents is given by

$$\tan^{-1}\left(\left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|\right) = \tan^{-1}\left(\frac{\sqrt{(m_1 + m_2)^2 - 4\,m_1 m_2}}{1 + m_1 m_2}\right) = \tan^{-1}\left(\frac{\sqrt{\frac{25}{4} - 4}}{1+1}\right) = \tan^{-1}\left(\frac{3}{4}\right).$$

Illustration 35

Prove that the tangents at the extremities of any focal chord intersect at right angles on the directrix.

Solution :

Let the parabola be $y^2 = 4ax$ ...(i)

Let $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$ be the extremities of a focal chord.

Equation of chord AB is $y(t_1 + t_2) = 2(x + at_1t_2)$ ...(ii)

Since line (ii) passes through the focus (a, 0)

$\therefore$ $2a(1 + t_1t_2) = 0$ or $t_1t_2 = -1$ ...(iii)

The equation of the tangents at A and B are respectively

$t_1y = x + at_1^2$ ...(iv)

and $t_2y = x + at_2^2$ ...(v)

Slope of line (iv) is $\frac{1}{t_1}$ and slope of line (v) is $\frac{1}{t_2}$.

Product of the slopes

$$= \frac{1}{t_1}.\frac{1}{t_2} = \frac{1}{t_1t_2} = \frac{1}{-1} = -1 \quad \text{[from (iii)]}$$

Hence the tangents intersect at right angles.

Also on solving (iv) and (v) we get $x = at_1t_2$

$\Rightarrow$ $x = -a$ which is independent of choice of t_1 and t_2, hence tangents will meet always on directrix, $x = -a$.

Illustration 36

Find the equation of the common tangents to the parabola $y^2 = 4ax$ and $x^2 = 4by$.

Solution :

The equation of any tangent in terms of slope (m) to the parabola $y^2 = 4ax$ is

$$y = mx + \frac{a}{m} \quad ...(1)$$

If this line is also tangent to the parabola $x^2 = 4ay$, then (1) meets $x^2 = 4by$ in two coincident points.

Substituting the value of y from (1) in $x^2 = 4by$ we get $x^2 = 4b\left(mx + \frac{a}{m}\right)$

$$x^2 - 4bmx - \frac{4ab}{m} = 0$$

The roots of this quadratic are equal provided "$B^2 = 4AC$"

i.e., $$(-4bm)^2 = 4.1\left(\frac{-4ab}{m}\right)$$

$\Rightarrow$ $$16b^2m^3 + 16ab = 0,\ m \neq 0$$

or $$m^3 = -a/b$$

$\therefore$ $$m = -a^{1/3}/b^{1/3}$$

Substituting the value of m in (1) the required equation is

$$y = -\frac{a^{1/3}}{b^{1/3}}x - \frac{ab^{1/3}}{a^{1/3}}$$

$\Rightarrow$ $$y = -\frac{a^{1/3}}{b^{1/3}}x - a^{2/3}b^{1/3}$$

$\Rightarrow$ $$a^{1/3}x + b^{1/3}y + a^{2/3}b^{2/3} = 0$$

Illustration 37

The tangents to the parabola $y^2 = 4ax$ make angle θ_1 and θ_2 with x-axis. Find the locus of their point of intersection if $\cot\theta_1 + \cot\theta_2 = c$.

Solution :

Let the equation of any tangent to the parabola $y^2 = 4ax$ is

$$y = mx + (a/m) \quad ...(1)$$

Let (x_1, y_1) be the point of intersection of the tangents to $y^2 = 4ax$ then (1) passes through (x_1, y_1)

$\therefore$ $$y_1 = mx_1 + (a/m)$$

or $$m^2x_1 - my_1 + a = 0$$

Let m_1 and m_2 be the roots of this quadratic equation then

$$m_1 + m_2 = y_1/x_1 \text{ and } m_1m_2 = a/x_1$$

or $$\tan\theta_1 + \tan\theta_2 = y_1/x_1$$

and $$\tan\theta_1 \tan\theta_2 = a/x_1 \quad ...(2)$$

Now $$\cot\theta_1 + \cot\theta_2 = c \text{ (given)}$$

$\Rightarrow$ $$\frac{1}{\tan\theta_1} + \frac{1}{\tan\theta_2} = c$$

$\Rightarrow$ $$\frac{\tan\theta_1 + \tan\theta_2}{\tan\theta_1 \tan\theta_2} = c \quad \text{[From (2)]}$$

$\Rightarrow \qquad \dfrac{y_1/x_1}{a/x_1} = c$

$\Rightarrow \qquad y_1 = ac$

The required locus is $\qquad y = ac \qquad$ (which is a line parallel to x-axis)

6.1 EQUATION OF NORMAL IN DIFFERENT FORMS

Type 1 : POINT FORM

The equation of normal to the parabola $y^2 = 4ax$ at a point $(x_1\ y_1)$ is

$$\boxed{y - y_1 = \frac{y_1}{2a}(x - x_1)}$$

logic : we need to find the slope of normal to find the equation. We earlier calculated the slope of tangent

i.e. $\qquad \left.\dfrac{dy}{dx}\right|_{x_1 y_1} = \dfrac{2a}{y_1}$

$\therefore \qquad$ Slope normal $= \dfrac{-1}{\text{Slope tangent}} = \dfrac{-y_1}{2a}$

as $m_1 m_2 = -1$ for two perpendicular lines.

Hence the equation of normal becomes

$$y - y_1 = \frac{y_1}{2a}(x - x_1)$$

which is the required equation.

Type 2 : PARAMETRIC FORM

The equation of normal to the parabola $y^2 = 4ax$ at parametric point $(at^2, 2at)$ is

$$\boxed{y + tx = 2at + at^3}$$

for proof just replace x_1 by at^2 and y by 2at in point form equation.

Type 3 : SLOPE FORM

The equation of normal of slope m to the parabola $y^2 = 4ax$ at the point $(am^2, -2am)$ (note that point is also in slope form) is

$$\boxed{y = mx - 2am - am^3}$$

from point form we know eq^n of normal is

$$y - y_1 = \frac{-y_1}{2a}(x - x_1) \qquad \text{...(i)}$$

$\Rightarrow \qquad m = \dfrac{-y_1}{2a} \Rightarrow y_1 = -2am \qquad$ (which is the ordinate of point taken)

for abcissa we know point lies on parabola

hence $$x_1 = \frac{y_1^2}{4a} = \frac{4a^2m^2}{4a} = am^2$$

hence here $(x_1\ y_1) = (am^2, -2am)$

put these values of x_1y_1 in (i) to get the required equation

$$y + 2am = m(x - am^2)$$

$$\Rightarrow \quad y = mx - 2am - am^3$$

6.2 CONDITION FOR A LINE TO BE A NORMAL

The condition required for a line $y = mx + c$ to be a normal to the parabola $y^2 = 4ax$ is

$$\boxed{c = -2am - am^3}$$

Logic : Nothing just compare the equation of normal derived just above in slope form with $y = mx + c$ to arrive at the result.

Point of intersection of the normals to the parabola $y^2 = 4ax$ at $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$:

Equation of normals at $A(at_1^2, 2at_1)$ and $B(at_2^2, at_2)$ are

$$y + t_1x = 2at_1 + at_1^3 \quad ...(1)$$

$$y + t_2x = 2at_2 + at_2^3 \quad ...(2)$$

$$(1) - (2) \Rightarrow (t_1 - t_2)x = 2a(t_1 - t_2) + a(t_1^3 - t_2^3)$$

$$\Rightarrow x = 2a + a(t_1^2 + t_2^2 + t_1t_2)$$

$$t_2 \cdot (1) - t_1(2) \Rightarrow (t_2 - t_1)y = at_1t_2(t_1^2 - t_2^2)$$

$$\Rightarrow y = -at_1t_2(t_1 + t_2)$$

Thus point of intersection of lines (1) and (2) is

$$\boxed{(2a + a(t_1^2 + t_2^2 + t_1t_2), -at_1t_2(t_1 + t_2))}$$

6.3 RELATION BETWEEN 't_1' AND 't_2' IF NORMAL AT 't_1' MEETS THE PARABOLA AGAIN AT 't_2'

Let the parabola be $y^2 = 4ax$, equation of normal at $P(at_1^2, 2at_1)$ is

$$y = -t_1x + 2at_1 + at_1^3 \quad ...(1)$$

Since it meet the parabola again at $Q(at_2^2, 2at_2)$ then equ

(1) passes through $Q(at_2^2, 2at_2)$

$$\therefore \quad 2at_2 = -at_1t_2^2 + 2at_1 + at_1^3$$

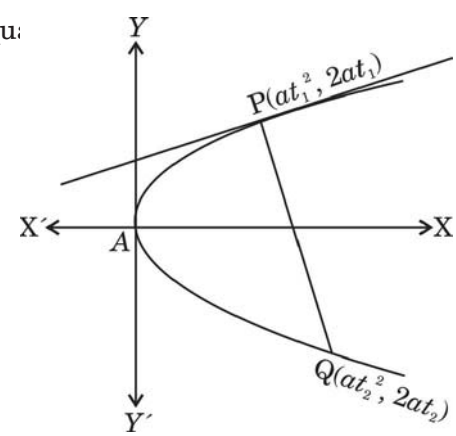
$$\Rightarrow \quad 2a(t_2 - t_1) + at_1(t_2^2 - t_1^2) = 0$$

$$\Rightarrow \quad a(t_2 - t_1)[2 + t_1(t_2 + t_1)] = 0$$

$$\because \quad a(t_2 - t_1) \neq 0 \quad (\because t_1 \text{ and } t_2 \text{ are different})$$

$$\therefore \quad 2 + t_1(t_2 + t_1) = 0$$

$$\therefore \quad \boxed{t_2 = -t_1 - \frac{2}{t_1}}$$



Result 2 : If normals at 't_1' and 't_2' meets the parabola $y^2 = 4ax$ at same point then

$$t_1t_2 = 2$$

Proof : Suppose normals meet at 'T' then

$$T = -t_1 - \frac{2}{t_1} = -t_2 - \frac{2}{t_2}$$

$$(t_1 - t_2) = 2\left(\frac{1}{t_2} - \frac{1}{t_1}\right)$$

$$\mathbf{t_1t_2 = 2} \qquad (\because t_1 \neq t_2)$$

Illustration 38

Find the equations of the normals at the ends of the latus-rectumof the parabola $y^2 = 4ax$. Also prove that they are at right angles on the axis of the parabola.

Solution :

The coordinates of the ends of the latusrectum of the parabola $y^2 = 4ax$ are (a, 2a) and (a, –2a) is

$$y - 2a = -\frac{2a}{2a}(x - a) \text{ or } x + y - 3a = 0 \qquad ...(i)$$

$$y + 2a = \left(\frac{-2a}{2a}\right)(x - a) \text{ or } x - y - 3a = 0 \qquad ...(ii)$$

Clearly, (i) and (ii) are perpendicular as the product of their slopes is – 1. The point of intersection of (i) and (ii) is (3a, 0), which lies onthe axis of the parabola.

Illustration 39

Prove that the chord $y - x\sqrt{2} + 4a\sqrt{2} = 0$ is a normal chord of the parabola $y^2 = 4ax$. Also, find the point on the parabola where the given chord is normal to the parabola.

Solution :

We have : $y - x\sqrt{2} + 4a\sqrt{2} = 0$

or $$y = \sqrt{2}\,x - 4a\sqrt{2} \qquad ...(i)$$

The equation of any normal to the parabola $y^2 = 4ax$ is

$$y = mx - 2am - am^3 \qquad ...(ii)$$

Comparing (i) and (ii), we get

$$m = \sqrt{2} \text{ and } -2am - am^3 = -4a\sqrt{2}$$

Clearly, $m = \sqrt{2}$ satisfies $-2am - am^3 = -4a\sqrt{2}$.

Hence, the given chord is normal to the parabola $y^2 = 4ax$.

The coordinates of the point are $(am^2, -2am)$ i.e. $(2a, -2\sqrt{2}\,a)$.

Illustration 40

Prove that the tangent at one extremity of the focal chord of a parabola is parallel to the normal at the other extremity.

Solution :

The PQ be a focal chord of the parabola, and let the coordinates of P and Q be $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ respectively. Then,

$$t_1 \; t_2 = -1 \qquad ...(i)$$

The equation of tangent at P $(at_1^2, 2at_1)$ to $y^2 = 4ax$ is

$$t_1 \; y = x + at_1^2 \qquad ...(ii)$$

The equation of the normal at $Q(at_2^2, 2at_2)$ to $y^2 = 4ax$ is

$$y + t_2x = 2at_2 + at_2^3 \qquad ...(iii)$$

The slope the tangent at P = is $m_1 = \frac{1}{t_1}$

The slope of the normal at Q is,

$$m_2 = -t_2 = \frac{1}{t_1} \qquad [\because \; t_1 \; t_2 = -1 \; \therefore \; t_2 = -1/t_1]$$

Clearly, $m_1 = m_2$. Hence, (i) and (ii) are parallel.

Illustration 41

If the normal at a point $P(at^2, 2at)$ to the parabola $y^2 = 4ax$ subtends a right angle at the vertex of the parabola. Then prove that $t^2 = 2$.

Solution :

The equation of the normal to the parabola $y^2 = 4ax$ at P is $y + tx = 2at + at^3$. Suppose it meets the parabola at Q. If O is the vertex of the parabola, then the combined equation of OPand OQ is a homogeneous equation of second degree, given by

$$y^2 = 4ax\left(\frac{y+tx}{2at+at^3}\right)$$

$\Rightarrow \qquad y^2(2at + at^3) = 4ax(y + tx)$

$\Rightarrow \qquad 4at\,x^2 + 4axy - (2at + at^3)\,y^2 = 0$

If OP and OQ are at right angle, then

Coeff. of x^2 + Coeff. of y^2 = 0

$\Rightarrow \qquad 4at - 2at - at^3 = 0 \Rightarrow t^2 = 2.$

Illustration 42

If the normals at two points P and Q of a parabola $y^2 = 4ax$ intersect at a third point R on the curve. Show that the product of the ordinates of P and Q is $8a^2$.

Solution :

Let P $(at_1^2, 2at_1)$ and Q$(at_2^2, 2at_2)$ be two points on the parabola $y^2 = 4ax$. It is given that the normals at P and Q intersect at R on the parabola. Therefore,

$$t_1 t_2 = 2.$$

So, product of the ordinates at P and Q $= (2at_1)(2at_2)$

$$= 4a^2 + t_1 t_2 = 8a^2 \qquad [\because t_1 t_2 = 2]$$

Illustration 43

Prove that the normal chord of a parabola at the point whose ordinate is equal to the abscissa subtends a right angle at the focus.

Solution :

Let $P(at_1^2, 2at_1)$ be a point on the parabola and PQ be a chord normal at point P. Let the coordinates of Q be $(at_2^2, 2at_2)$. Then,

$$t_2 = -t_1 - \frac{2}{t_1}$$

It is given that $at_1^2 = 2at_1 \Rightarrow t_1 = 0$ or $t_1 = 2$.

For $t_1 = 0$, t_2 is not defined, which means that the normal chord does not intersect the parabola.

$$\therefore \qquad t_1 = 2 \Rightarrow t_2 = -2 - \frac{2}{2} = -3.$$

Thus, the coordinates of P and Q are (4a, 4a) and (9a, – 6a). Let S (a, 0) be the focus of the parabola.

Thus, $$m_1 = \text{slope of SQ} = \frac{9a - a}{-6a - 0} = -\frac{4}{3}$$

and, $$m_2 = \text{slope of SP} = \frac{4a - a}{4a - 0} = \frac{3}{4}$$

Clearly, $$m_1 m_2 = -1 \Rightarrow SP \perp SQ.$$

Hence, PQ subtends a right angle at S.

Illustration 44

Prove that two parabolas $y^2 = 4ax$ and $y^2 = 4c\ (x - b)$ cannot have a common normal unless $\frac{b}{a-c} > 2$.

Solution :

The equation of any normal of slope m to the parabola $y^2 = 4c\ (x - b)$ is

$$y = m\ (x - b) - 2cm - cm^3$$

or, $$y = mx - mb - 2cm - cm^3$$

For this to be normal to $y^2 = 4ax$, we must have

$$-mb - 2cm - cm^3 = -2am - am^3$$

$$\Rightarrow \quad b + 2c + cm^2 = 2a + am^2$$

$$\Rightarrow \quad b + 2c - 2a = (a - c)\ m^2$$

$$\Rightarrow \quad m^2 = \frac{b - 2(a-c)}{a-c}$$

$$\Rightarrow \quad m = \sqrt{\frac{b}{a-c} - 2}$$

Since m is real, therefore,

$$\frac{b}{a-c} - 2 > 0 \Rightarrow \frac{b}{a-c} > 2.$$

Illustration 45

If normal at the point $(at^2, 2at)$ in the parabola $y^2 = 4ax$ intersects the parabola again at the point $(am^2, 2am)$, then find the minimum value of m^2.

Solution :

Since normal at the point 't' intersects again at the point 'm'

we have $m = -t - \frac{2}{t}$

$$\Rightarrow \quad t^2 + tm + 2 = 0$$

since 't' is real, $D \geq 0$

$m^2 \geq 8$

so, least value of m^2 is 8.

6.4 FROM ANY POINT THREE NORMALS CAN BE DRAWN TO A PARABOLA OF WHICH AT LEAST ONE WILL BE REAL.

Let the equation of the parabola be $y^2 = 4ax$ and let the given point be (x_1, y_1).

The equation of any normal to the parabola in the slope form is

$$y = mx - 2am - am^3 \quad ...(1)$$

If this normal passes through (x_1, y_1), then

$$y_1 = mx_1 - 2am - am^3 \text{ or } am^3 + (2a - x_1)m + y_1 = 0.$$

This is a cubic equation in m and has three roots, say m_1, m_2 and m_3.

Corresponding to these three values of m we will get three normals to the parabola passing through (x_1, y_1).

Since imaginary roots of a polynomial equation with real coefficients always occur in pair, therefore at least one root will be real and hence at least one normal will be real.

Note : Any three points on a parabola normals at which pass through a common point are called co-normal points.

Illustration 46

Three normals are drawn from the point (c, 0) to the curve $y^2 = x$. Show that c must be greater than 1/2. One normal is always the x-axis. Find c for which the other two normals are perpendicular to each other.

Solution :

Equation to normal to the parabola

$$y^2 = x \text{ is } y = mx - \frac{m}{2} - \frac{m^3}{4} \text{ is passing through } (c, 0)$$

$$0 = cm - \frac{m}{2} - \frac{m^3}{4} \Rightarrow m\left(c - \frac{1}{2}\right) = \frac{m^3}{4}$$

$$m = 0 \text{ or } c - \frac{1}{2} = \frac{m^2}{4}$$

$$c - \frac{1}{2} \geq 0 \left(\text{since } \frac{m^2}{4} \geq 0\right)$$

$$c \geq \frac{1}{2}. \text{ If } c = \frac{1}{2}, \text{ then } m = 0$$

Then only one normal will be there i.e., x-axis.

Since three normals are there, $\therefore c > \frac{1}{2}$

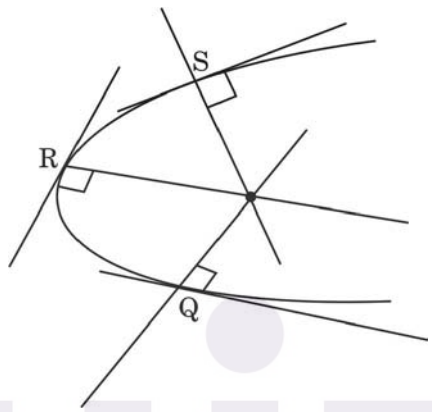
Then $$\left(2\sqrt{c-\frac{1}{2}}\right)\left(-2\sqrt{c-\frac{1}{2}}\right) = -1$$

$\Rightarrow$ $$c - \frac{1}{2} - \frac{1}{4} \Rightarrow c = \frac{3}{4}$$

6.5 CO-NORMAL POINTS

The points on the curve at which the normals pass through a common point are called co-normal points.

Q, R and S are co-normal points. The co-normal points are also called the feet of the normals.



SOME RESULTS ON CO-NORMAL POINTS

Let $y^2 = 4ax$ be a parabola and let P (x_1, y_1) a point. The equation of any normal to $y^2 = 4ax$ is

$$y = mx - 2am - am^3$$

If it passes thought $P(x_1, y_1)$, then

$$y_1 = mx_1 - 2am - am^3$$

$\Rightarrow$ $$am^3 + (2a - x_1) + y_1 = 0$$

This is a cubic equation in m. So, it gives three values of m, say, m_1, m_2, m_3. Corresponding to each of these values of m there is a normal to the parabola passing through point $P(x_1, y_1)$. Let Q, R and S be three points on the parabola such that the normals at these points pass through P. If m_1, m_2 and m_3 are the slopes of the normals at Q, R and S respectively. Then, the coordinates of Q, R and S are $(am_1^2, -2am_1)$, $(am_2^2, -2am_2)$ and $(am_3^2, -2am_3)$ respectively. These three points are called the feet of the normals.

Now, m_1, m_2 and m_3 are roots of (i). Therefore,

$$m_1 + m_2 + m_3 = 0 \quad \text{...(ii)}$$

$$m_1 m_2 + m_2 m_3 + m_3 m_1 = \frac{(2a - x_1)}{a} \quad \text{...(iii)}$$

and, $$m_1 m_2 m_3 = -\frac{y_1}{a} \quad \text{...(iv)}$$

Result I. The sum of the slopes of the normals at conormal points is zero.

Result II. The sum of the ordinates of the co-normal points is zero.

Proof : The coordinates of the feet of the normals are $(am_1^2, -2am_1)$, $(am_2^2, -2am_2)$ and $(am_3^2, -2am_3)$.

$\therefore$ Sum of the ordinates of the co-normal points

$$= -2a(m_1 + m_2 + m_3) = -2a \quad 0 \qquad \text{[Using (i)]}$$

$$= 0$$

Result III. The centroid of the triangle formed by the co-normalpoints lies on the axis of the parabola :

Proof : The vertices of the triangle formed by the co-normal points are $(am_1^2, -2am_1)$, $(am_2^2, -2am_2)$ and $(am_3^2, -2am_3)$.

So, y-coordinates of the centroid $= -\frac{2a}{3}(m_1 + m_2 + m_3)$

$$= -\frac{2a}{3} \quad 0 = 0$$

Hence, the centroid lies on the x-axis i.e. axis of the parabola.

Illustration 47

Find the locus of the points of intersection of the three normals to the parabola $y^2 = 4ax$, two of which are inclined at right angles to each other.

Solution :

Let (h, k) be the point of intersection of three normals to the parabola $y^2 = 4ax$. The equation of any normal to $y^2 = 4ax$ is

$$y = mx - 2am - am^3.$$

If it passes through (h, k), them

$$k = mh - 2am - am^3$$

$$\Rightarrow \quad am^3 + m(2a - h) + k = 0 \qquad \text{...(i)}$$

This equation, being a cubic in m, gives three values of m say, m_1, m_2, m_3.

$$\therefore \quad m_1, m_2, m_3 = -k/a$$

But two of the three normals are at right angles.

$$\therefore \quad m_1 m_2 = -1 \Rightarrow -m_3 = -\frac{k}{a} \Rightarrow m_3 = \frac{k}{a}.$$

Since m_3 is a root of (i), therefore,

$$am_3^3 + m_3(2a - h) + k = 0$$

$$\Rightarrow \quad a\left(-\frac{k}{a}\right)^3 - \frac{k}{a}(2a - h) + k = 0 \qquad [\because m_3 = -k/a]$$

$\Rightarrow \qquad k\{k^2 + (3a - h)a\} = 0$

So, locus of (h, k) is $y(y^2 + (3a - h)a) = 0$

Illustration 48

Prove that the locus of points such that two of the three normals from them to the parabola $y^2 = 4ax$ coincide is $27ay^2 = 4(x - 2a)^3$.

Solution :

we have

$$m_1 + m_2 + m_3 = 0 \qquad ...(i)$$

$$m_1 m_2 + m_2 m_3 + m_3 m_1 = \frac{(2a - h)}{a} \qquad ...(ii)$$

and,
$$m_1 m_2 m_3 = -\frac{k}{a} \qquad ...(iii)$$

But here two of the three normals are given to be coincident i.e. $m_1 = m_2$

Putting $m_1 = m_2$ in (i) and (iii), we get

$$2m_1 + m_3 = 0 \qquad ...(iv)$$

and,
$$m_1^2 m_3 = \frac{-k}{a} \qquad ...(v)$$

Putting $m_3 = -2m_1$, from (iv) in (v), we get

$$-2m_1^3 = -\frac{k}{a} \Rightarrow m_1^3 = \frac{k}{2a}$$

Since m_1 is a root of the equation

$$am^3 + m(2a - h) + k = 0$$

$\therefore \qquad am_1^3 + m_1(2a - h) + k = 0$

$\Rightarrow \qquad \frac{k}{2} + \left(\frac{k}{2a}\right)^{1/3}(2a - h) + k = 0 \qquad$ [Putting $m_1 = (k/2a)^{1/3}$

$\Rightarrow \qquad \left(\frac{k}{2a}\right)^{1/3}(2a - h) = -\frac{3k}{2}$

$$\frac{k}{2a}(2a - h)^3 = \frac{-27k^3}{8} \Rightarrow 27ak^2 = 4(h - 2a)^3$$

Hence, the locus of (h, k) is

$$27ay^2 = 4(x - 2a)^3$$

Illustration 49

Find the locus of a point which is such that (α) two of the normals drawn from it to the parabola are at right angles, (β) the three normals through it cut the axis in points whose distances from the vertex are in arithmetical progression.

Solution :

Any normal is $y = mx - 2am - am^3$, and this passes through the point (h, k), if

$$am^3 + (2a - h)m + k = 0 \quad ...(1)$$

If then m_1, m_2 and m_3 be the roots, we have, by Art.2,

$$m_1 + m_2 + m_3 = 0 \quad ...(2)$$

$$m_2m_3 + m_3m_1 + m_1m_2 = \frac{2a-h}{a} \quad ...(3)$$

and

$$m_1m_2m_3 = -\frac{k}{a} \quad ...(4)$$

(a) If two of the normals, say m_1 and m_2, be at right angles, we have $m_1m_2 = -1$, and hence, from (4), $m_3 = \frac{k}{a}$

The quantity $\frac{k}{a}$ is therefore, a root of (1) and hence, by substitution, we have

$$\frac{k^3}{a^2} + (2a - h)\frac{k}{a} + k = 0$$

i.e.,

$$k^2 = a(h - 3a)$$

The locus of the point (h, k) is therefore the parabola $y^2 = a(x - 3a)$ whose vertex is the point (3a, 0) and whose latus rectum is one-quarter that of the given parabola.

The student should draw the figure of both parabolas.

(β) The normal $y = mx - 2am - am^3$ meets the axis of x at a point whose distance fromthe vertex is $2a + am^2$. The conditions of the question then give

$$(2a + am_1^2) + (2a + am_3^2) = 2(2a + am_2^2)$$

i.e.,

$$m_1^2 + m_3^2 = 2m_2^2 \quad ...(5)$$

If we eliminate m_1, m_2 and m_3 from the equations (2), (3), (4) and (5), we shall have a relation between h and k.

From (2) and (3), we have

$$\frac{2a-h}{a} = m_1m_3 + m_2(m_1 + m_3) = m_1m_3 - m_2^2 \qquad ...(6)$$

Also, (5) and (2) give

$$2m_2^2 = (m_1 + m_3)^2 - 2m_1m_3 = m_2^2 - 2m_1m_3$$

i.e., $$m_2^2 + 2m_1m_3 = 0 \qquad ...(7)$$

Solving (6) and (7), we have

$$m_1m_3 = \frac{2a-h}{3a}, \text{ and } m_2^2 = -2\ \frac{2a-h}{3a}$$

Substituting these values in(4), we have

$$\frac{2a-h}{3a}\sqrt{-2\frac{2a-h}{3a}} = -\frac{k}{a}$$

i.e., $$27ak^2 = 2(h - 2a)^3$$

so that the required locus is

$$27ay^2 = 2(x - 2a)^3$$

Illustration 50

If the normals at three points P, Q **and** R **meet in a point** O **and** S **be the focus, prove that** SP . SQ . SR = a . SO^2**.**

Solution :

As in the previous question we know that the normals at the points $(am_1^2, -2am_1)$, $(am_2^2, -2am_2)$ and $(am_3^2, -2am_3)$ meet in the point (h, k), if

$$m_1 + m_2 + m_3 = 0 \qquad ...(1)$$

$$m_2m_3 + m_3m_1 + m_1m_2 = \frac{2a-h}{a} \qquad ...(2)$$

and $$m_1m_2m_3 = -\frac{k}{a} \qquad ...(3)$$

By Art. 202, we have

$$SP = a(1 + m_1^2),\ SQ = a\ (1 + m_2^2), \text{ and } SR = a\ (1 + m_3^2)$$

Hence, $$\frac{SP.SQ.SR}{a^3} = (1 + m_1^2)\ (1 + m_2^2)\ (1 + m_3^2)$$

$$= 1 + (m_1^2 + m_2^2 + m_3^2) + (m_2^2\ m_3^2 + m_3^2\ m_1^2 + m_1^2\ m_2^2) + m_1^2\ m_2^2\ m_3^2$$

Also, from (1) and (2), we have

$$m_1^2 + m_2^2 + m_3^2 = (m_1 + m_2 + m_3)^2 - 2(m_2m_3 + m_3m_1 + m_1m_2)$$

$$= 2\left(\frac{h-2a}{a}\right)$$

and $$m_2^2\, m_3^2 + m_3^2\, m_1^2\, m_2^2 = (m_2m_3 + m_3m_1 + m_1m_2)^2 - 2m_1m_2m_3\,(m_1 + m_2 + m_3)$$

$$=\left(\frac{h-2a}{a}\right)^2 \qquad \text{by (1) and (2)}$$

Hence, $$\frac{SP.SQ.SR}{a^3} = 1 + 2\frac{h-2a}{a} + \left(\frac{h-2a}{a}\right)^2 + \frac{k^2}{a^2}$$

$$=\frac{(h-a)^2 + k^2}{a^2} = \frac{SO^2}{a^2}$$

i.e., $$SP\,.\,SQ\,.SR = SO^2\,.\,a$$

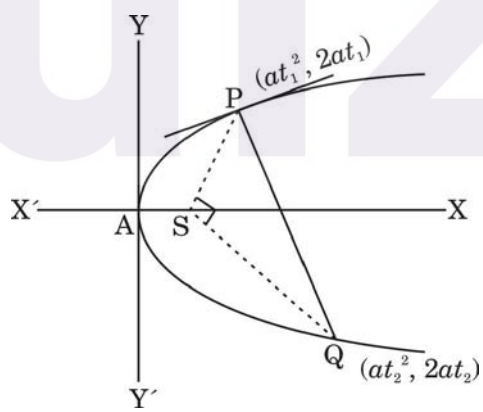
Illustration 51

Prove that the normal chord to a parabola $y^2 = 4ax$ at the point whose ordinate is equal to abscissa subtends a right angle at the focus.

Solution :

Let the normal at $P(at_1^2, 2at_1)$ meet the curve at $Q(at_2^2, 2at_2)$ PQ is a normal chord.

and $$t_2 = -\,t_1 - \frac{2}{t_1} \qquad ...(1)$$



By given condition $$2at_1 = at_1^2$$

$\therefore$ $t_1 = 2$ from equation (1), $t_2 = -\,3$

then $P(4a, 4a)$ and $Q(9a, -\,6a)$

but focus $S(a, 0)$

$\therefore$ $$\text{Slope of SP} = \frac{4a-0}{4a-a}$$

$$= \frac{4a}{3a} = \frac{4}{3}$$

and $$\text{Slope of SQ} = \frac{-6a - 0}{9a - a}$$

$$= -\frac{6a}{8a} = -\frac{3}{4}$$

. $$\text{Slope of SP} \quad \text{Slope of SQ} = \frac{4}{3} \times -\frac{3}{4} = -1$$

$\therefore$ $$\angle PSQ = \pi/2$$

i.e., PQ subtends a right angle at the focus S.

Illustration 52

Three normals are drawn from the point (7, 14) to the parabola $x^2 - 8x - 16y = 0$. Find the coordinates of the feet of the normals.

Solution :

The equation of the given parabola is

$$x^2 - 8x - 16y = 0$$

Differentiating above equation thrughout w.r.t. x, we have

$$2x - 8 - 16\frac{dy}{dx} = 0$$

gives $$\frac{dy}{dx} = \frac{x-4}{8}$$

If (h, k) be a point on the given parabola, then we have

$$k = \frac{h^2 - 8h}{16}$$

and slope of the normal at (h, k) $= \dfrac{8}{4-h}$

Therefore, equation of the normal at (h, k) is

$$y - k = \left(\frac{8}{4-h}\right)(x - h)$$

If the normal passes through (7, 14), then we have

$$14 - \frac{h^2 - 8h}{16} = \left(\frac{8}{4-h}\right)(7 - h)$$

i.e., $(4 - h)(224 - h^2 + 8h) = 128(7 - h)$

i.e., $h^3 - 12h^2 - 64h = 0$

i.e., $h(h^2 - 12h - 64) = 0$

gives $h = 0, -4, 16$

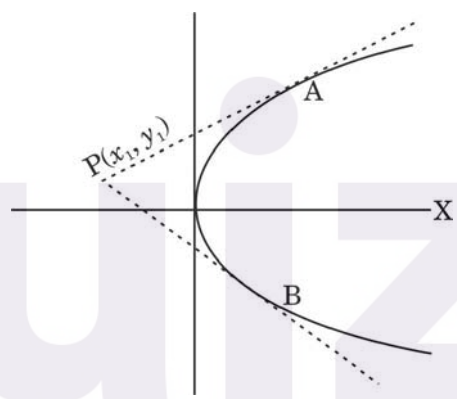
Putting the values of h in equation (3) gives k = 0, 3, 8 respectively.

Therefore, the coordinates of the feet of the normals are (0, 0), (– 4, 3) and (16, 8).

7.1 THE EQUATION OF CHORD OF CONTACT

If from $P(x_1, y_1)$ tangents PA and PB are drawn to the parabola $y^2 = 4ax$ then equation of chord of contact AB is $\boxed{yy_1 = 2a(x + x_1)}$

which follows from the fact that the equation which is being claimed to be equation of AB represents a line and is satisfied by co-ordinates of A and B. Indeed if A be (α, β) then AP is tangent at P



$\Rightarrow$ Equation of AP is $y\beta = 2a(x + \alpha)$

Since it passes through $P(x_1, y_1)$ we must have

$$\beta y_1 = 2a(x_1 + \alpha)$$

which shows that the point $A(\alpha, \beta)$ lies on the line

$$yy_1 = 2a(x + x_1)$$

Similarly the point B lies on the line $yy_1 = 2a(x + x_1)$ which completes the proof.

NOTE :

The equation of chord of contact resembles with that of the tangent's equation but in the tangent's case the point (x_1, y_1) lies on the parabola.

Illustration 53

Find the equation of the chord of contact of tangents drawn from a point (3, 4) to the parabola $y^2 = 2x$.

Solution :

Using $yy_1 = 2a(x + x_1)$, the equation of the chord of contact is

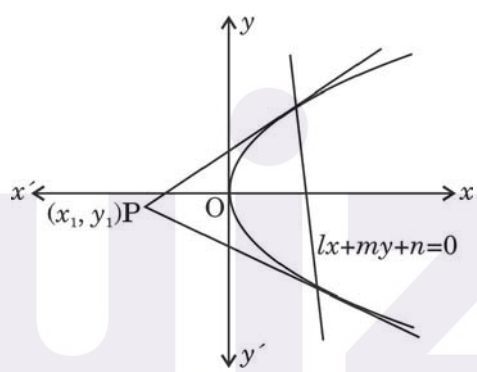
$$4y = (x + 3) \text{ or } x - 4y + 3 = 0$$

Illustration 54

Tangents are drawn at the points where the line $lx + my + n = 0$ is intersected by the parabola $y^2 = 4ax$. Find the point of intersection of tangents.

Solution :

Let the point of intersection of tangents be (x_1, y_1) then the equation of the chord of contact of tangents drawn from P to the parabola $y^2 = 4ax$ is



$$yy_1 = 2a(x + x_1)$$

Clearly, $lx + my + n = 0$ is also the chord of contact of tangents. Therefore, $yy_1 = 2a(x + x_1)$ and $lx + my + n = 0$ represent the same line. Hence,

$$\frac{2a}{l} = \frac{-y_1}{m} = \frac{2a\ x_1}{n}$$

$$\Rightarrow \qquad x_1 = \frac{n}{l} \text{ and } y_1 = \frac{-2am}{l}$$

Hence, the required point is $\left(\frac{n}{l}, \frac{-2am}{l}\right)$

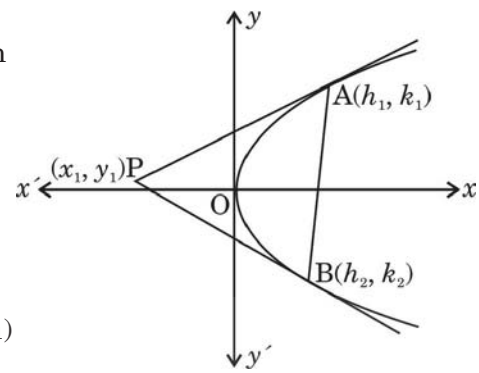
Illustration 55

Show that the length of the chord of contact of tangents drawn from (x_1, y_1) tothe parabola $y^2 = 4ax$ is

$$\frac{1}{a}\sqrt{(y_1^2 - 4ax_1)(y_1^2 + 4a^2)}$$

Solution :

Let AB be the chord of contact of tangents drawn from a point $P(x_1, y_1)$ to the parabola $y^2 = 4ax$. Then, the equation of AB is



$$yy_1 = 2a(x + x_1)$$

The ordinates of A and B are the roots of the equation

$$y^2 = 4a\left(\frac{yy_1 - 2ax_1}{2a}\right) \text{ or } y^2 - 2yy_1 + 4ax_1 = 0$$

$\therefore \quad k_1 + k_2 = 2y_1$ and $k_1 k_2 = 4ax_1$ ...(i)

Illustration 56

Prove that the area of the triangle formed by the tangents drawn from (x_1, y_1) to $y^2 = 4ax$ and their chord of contact is $\frac{(y_1^2 - 4ax_1)^{3/2}}{2a}$.

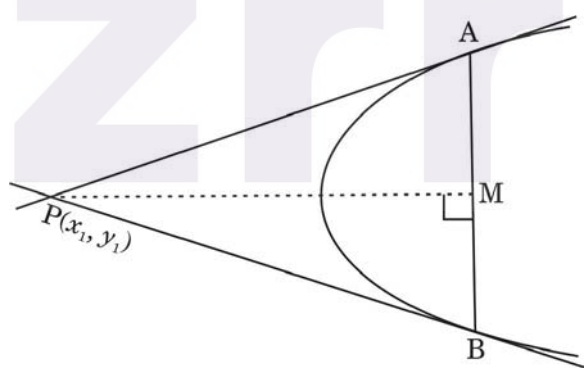
Solution :

Equation of AB (chord of contact) is

$$yy_1 = 2a(x + x_1)$$

or $\quad yy_1 - 2a(x + x_1) = 0$

. PM = length of perpendicular from $P(x_1, y_1)$ on AB



$$= \frac{|y_1 y_1 - 2a(x_1 + x_1)|}{\sqrt{(y_1^2 + 4a^2)}}$$

$$= \frac{(y_1^2 - 4ax_1)}{\sqrt{y_1^2 + 4a^2)}}$$

(since $P(x_1, y_1)$ lies outside the parabola $\therefore y_1^2 - 4ax_1 > 0$} and area of $\Delta PAB = \frac{1}{2}AB.PM$

$$= \frac{1}{2}.\frac{1}{|a|}.\sqrt{(y_1^2 - 4ax_1)(y_1^2 + 4a^2)}\ .\ \frac{(y_1^2 - 4ax_1)}{\sqrt{(y_1^2 + 4a^2)}}$$

$$= \frac{(y_1^2 - 4ax_1)^{3/2}}{2a}, \text{ if } a > 0.$$

8.1 EQUATION OF THE CHORD OF THE PARABOLA WHICH IS BISECTED AT A GIVE POINT

The equation of the chord of the parabola $y^2 = 4ax$ which is bisected at (x_1, y_1) is

$$\mathbf{yy_1 - 2a\ (x + x_1) = y_1^2 - 4a\,x_1 \text{ or } \boxed{T = S'}}$$

where $S' = y_1^2 - 4ax_1$ and $T = yy_1 - 2a\ (x + x_1)$.

Proof : The equation of a line passing through (x_1, y_1) is

$$y - y_1 = m\ (x - x_1) \qquad ...(i)$$

where m is a variable.

The abscissae of the points of intersection of $y^2 = 4ax$ and (ii) are the roots of the equation

$$(y_1 - mx_1 + mx)^2 = 4ax$$

or $\quad m^2\ x^2 + 2\ \{-\ 2a + my_1 - m^2\ x_1\}\ x + (y_1 - mx_1)^2 = 0$

Let p_1 and p_2 be the roots of this equation. Then,

$$p_1 + q_2 = \frac{-2\ \{-2\,a + my_1 - m^2 x_1\}}{m^2}$$

Since (x_1, y_1) is the mid-point of the chord joining the points of intersection of (i) and the parabola.

$$\therefore \qquad x_1 = \frac{p_1 + q_2}{2} \Rightarrow x_1 = \frac{-\{-2\,a + my_1 - m^2 x_1\}}{m^2} \Rightarrow m = \frac{2\,a}{y_1}$$

Substituting this value of m in (i), we obtain

$$y - y_1 = \frac{2\,a}{y_1}(x - x_1) \Rightarrow yy_1 - y_1^2 = 2a\ (x - x_1)$$

$$\Rightarrow \qquad yy_1 - 2a\ (x + x_1) = y_1^2 - 4ax_1 \Rightarrow T = S'$$

This is the equation of the chord bisected at (x_1, y_1).

Illustration 57

Find the equation of the chord of $y^2 = 8x$ which is bisected at (2, –3).

Solution :

The equation of the required chord is given by

$$S' = T \Rightarrow 9 - 8 \quad 2 = -\ 3y - 4\ (x + 2)$$

$$\Rightarrow 9 - 16 = -\ 4x - 3y - 8 \Rightarrow 4x + 3y + 1 = 0$$

Illustration 58

Find the mid-point of the chord intercepted on the line $4x - 3y + 4 = 0$ by the parabola $y^2 = 8x$.

Solution :

Let (x_1, y_1) be the middle point of the chord intercepted on the line $4x - 3y + 4 = 0$ by the parabola $y^2 = 8x$. Then the equation of the chord whose middle point is (x_1, y_1) is

$$yy_1 - 4(x + x_1) = y_1^2 - 8x_1$$

or, $$4x - yy_1 + y_1^2 - 4x_1 = 0 \quad ...(i)$$

Clearly, (i) and $4x - 3y + 4 = 0$ represent the same line.

$$\therefore \quad \frac{4}{4} = \frac{-y_1}{-3} = \frac{y_1^2 - 4x_1}{4} \Rightarrow 1 = \frac{y_1}{3} = \frac{y_1^2 - 4x_1}{4}$$

$\Rightarrow \quad y_1 = 3$ and $y_1^2 - 4x_1 = 4 \Rightarrow y_1 = 3$ and $x_1 = 5/4$

Hence, the required co-ordinates of the mid-point are (5/4, 3)

Illustration 59

Find the locus of the middle points of the chords of the parabola $y^2 = 4ax$ which pass through the focus.

Solution :

Let (h, k) be the co-ordinates the mid-point of one of the chords which pass through the focus (a, 0).

Then the equation of the chord whose mid-point is (h, k) is

$$ky - 2a(x + h) = k^2 - 4ah$$

or $$ky - 2ax + 2ah - k^2 = 0$$

If this line passes through the focus (a, 0), then

$$-2a^2 + 2ah - k^2 = 0 \Rightarrow k^2 - 2ah + 2a^2 = 0$$

Illustration 60

Find the locus of the mid points of the chords of the parabola $y^2 = 4ax$ which subtend a right angle at the vertex of the parabola.

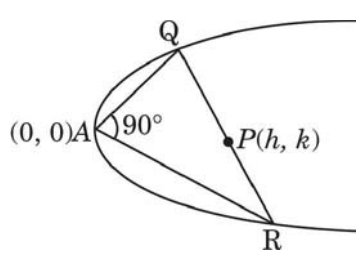
Solution :

Let P(h, k) be the mid point of a chord QR of the parabola $y^2 = 4ax$ then equation of chord QR is

$$T = S_1$$

or $$yk - 2a(x + h) = k^2 - 4ah$$

$\Rightarrow$ $$yk - 2ax = k^2 - 2ah \quad ...(1)$$



If A is the vertex of the parabola. For combined equation of AQ and AR making homogeneous of $y^2 = 4ax$ with the help of (1)

$$\therefore \quad y^2 = 4ax \ (1)$$

$$y^2 = 4ax\left(\frac{yk - 2ax}{k^2 - 2ah}\right)$$

$$y^2 (k^2 - 2ah) - 4akxy + 8a^2x^2 = 0$$

$\therefore$ Co-efficient of x^2 + Co-efficient of $y^2 = 0$

$$k^2 - 2ah + 8a^2 = 0$$

Hence the locus of P(h, k) is

$$y^2 - 2ax + 8a^2 = 0$$

Illustration 61

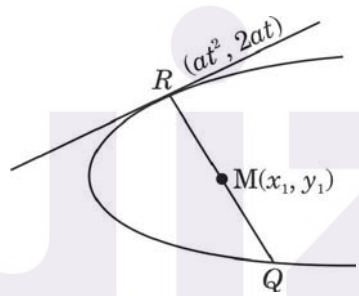
Show that the locus of the middle points of normal chords of the parabola $y^2 = 4ax$ is

$$y^4 - 2a\,(x - 2a)\,y^2 + 8a^4 = 0.$$

Solution :

Equation of the normal chord at any point $(at^2, 2at)$ of the parabola $y^2 = 4ax$ is

$$y + tx = 2at + at^3 \qquad ...(1)$$



But if $M(x_1, y_1)$ be its middle point its equation must be also

$$T = S_1$$

$$yy_1 - 2a\,(x + x_1) = y_1^2 - 4ax_1$$

or $$yy_1 - 2ax = y_1^2 - 2ax_1 \qquad ...(2)$$

. Equations (1) and (2) are identical, comparing them

$$\frac{1}{y_1} = \frac{t}{-2a} = \frac{2at + at^3}{y_1^2 - 2ax_1}$$

From first two relations $$t = -\frac{2a}{y_1} \qquad ...(3)$$

From last two relations $\frac{t}{-2a} = \frac{2at + at^3}{y_1^2 - 2ax_1}$

or $\frac{y_1^2 - 2ax_1}{-2a} = 2a + at^2$

or $\frac{y_1^2 - 2ax_1}{-2a} = 2a + a\left(\frac{-2a}{y_1}\right)^2$ (from equation (3)

or $\frac{y_1^2 - 2ax_1}{-2a} = \frac{2ay_1^2 + 4a^3}{y_1^2}$

or $y_1^4 - 2ax_1y_1^2 = -4a^2y_1^2 - 8a^4$

or $y_1^4 - 2a(x_1 - 2a)y_1^2 + 8a^4 = 0$

Hence locus of middle point (x_1, y_1) is

$$y^4 - 2a(x - 2a)y^2 + 8a^4 = 0$$

Illustration 62

If 't_1' and 't_2' are two points of the parabola and chord joining them subtend 90 at origin show that $t_1t_2 = -4$.

Solution :

The co-ordinate of points 't_1' and 't_2' are A $(at_1^2, 2at_1)$ and B $(at_2^2, 2at_2)$

$\Rightarrow$ Slope OA $= \frac{2at_1 - 0}{at_1^2 - 0} = \frac{2}{t_1}$

Similarly slope OB $= \frac{2}{t_2}$

We must have $\frac{2}{t_1} \cdot \frac{2}{t_2} = -1$

$\Rightarrow$ $t_1t_2 = -4$

Illustration 63

Find the locus of a point P which lies on the focal chord AB of the parabola $y^2 = 8x$ such that AP : PB = 2 : 1.

Solution :

Let A be $(at_1^2, 2at_1)$ and B be $(at_2^2, 2at_2)$ (Note that a = 2)

If P be the point (h, k) then $h = \frac{2at_2^2 + at_1^2}{3}$, $k = \frac{2.2at_2 + 2at_1}{3}$

and $t_1t_2 = -1$ (since AB is a focal chord)

The required locus must be eliminant of above equations. On putting a = 2 we have

$$3h = 4t_2^2 + 2t_1^2$$
$$3k = 8t_2 + 4t_1$$
$$t_2 = -\frac{1}{t_1}$$

On putting $t_2 = -\frac{1}{t_1}$ in the above system we get

$$\frac{4}{t_1^2} + 2t_1^2 = 3h \qquad (*)$$

$$4t_1 - \frac{8}{t_1} = 3k \qquad (**)$$

On squaring the second relation (**) we get

$$16t_1^2 + \frac{64}{t_1^2} - 64 = 9k^2 \qquad (***)$$

Multiplying the first relation by 64 and subtracting it from (***) (Eliminating $\frac{64}{t_1^2}$) we get

$$t_1^2 = \frac{48h - 9k^2 - 64}{9}$$

On putting (t_1^2 in (*) we get

$$\frac{36}{48h - 9k^2 - 64} + \frac{96h - 18k^2 - 128}{9} = 3h$$

Thus the locus of point P is

$$\frac{36}{48x - 9y^2 - 64} + \frac{96x - 18y^2 - 128}{9} = 3x$$

Illustration 64

Find the locus of point of intersection of tangents drawn at the extremities of normal chords of the parabola $y^2 = 8(x - 1)$.

Solution :

Putting x – 1 = X, y = Y the parabola becomes

$$Y^2 = 8X \text{ with } a = 2$$

Let the extremities of a normal chord be $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ then tangents at these points will intersect at a point (α, β) where

$$\alpha = at_1t_2,\ \beta = a(t_1 + t_2)$$

Also, $$t_2 = -t_1 - \frac{2}{t_1}$$

Putting a = 2 these relations become

$$\alpha = 2t_1t_2,\ \beta = 2(t_1 + t_2),\ t_2 = -t_1 - \frac{2}{t_1}$$

since we have to find the locus of (α, β) we must eliminate t_1 and t_2 from above equations. From third relation $t_1 + t_2 = -\frac{2}{t_1}$

and from second $$t_1 + t_2 = \frac{\beta}{2} \Rightarrow -\frac{2}{t_1} = \frac{\beta}{2} \Rightarrow t_1 = -\frac{4}{\beta}$$

whence $$t_2 = -t_1 - \frac{2}{t_1} = \frac{4}{\beta} + \frac{\beta}{2}$$

On substituting t_1, t_2 in first relation we get

$$\alpha = 2\left(-\frac{4}{\beta}\right)\left(\frac{4}{\beta} + \frac{\beta}{2}\right) \Rightarrow \beta(\alpha + 4) = -32$$

$\Rightarrow$ Locus of (α, β) is $Y^2(X + 4) = -32$

Switching over to old system of co-ordinates the locus is $y^2(x + 3) + 32 = 0$

9. POLE AND POLAR

If through a point P(within or outside a Parabola) there be drawn any straight line to meet the parabola in Q & R, then the locus of point of intersection of the tangents at Q & R is called the Polar of P; also P is called the Pole of the Polar.

The polar of a point (x_1y_1) with respect to the parabola $y^2 = 4ax$ is

$$\boxed{yy_1 = 2a(x + x_1)}$$

Note :

- When the point (x_1y_1) lies outside the parabola the equation to its polar is the same as the equation to the chord of contact of tangents drawn at (x_1y_1).
- When (x_1y_1) is on the parabola the polar is the same as the tangent at the point.

If the polar of a point P passes through the point T, then the polar of T goes through P.

Let P be the point (x_1, y_1) and T the point (h, k).

The polar of P is $yy_1 = 2a\ (x + x_1)$.

Since, it passes, through T, we have

$$y_1k = 2a(x_1 + h) \quad ...(1)$$

The polar of T is, $yk = 2a\ (x + h)$.

Since, (1) is true, this equation is satisfied by the coordinates x_1 and y_1.

Hence, the proposition.

Cor. : The point of intersection, T, of the polars of two points, P and Q, is the pole of the line PQ.

To find the pole of a given straight line with respect to the parabola.

Let the given straight line be

$$Ax + By + C = 0$$

If its pole be the point (x_1, y_1), it must be the same straight line as,

$$yy_1 = 2a(x + x_1)$$

i.e., $2ax - yy_1 + 2ax_1 = 0$

Since, these straight lines are the same, we have

$$\frac{2a}{A} = \frac{-y_1}{B} = \frac{2ax_1}{C}$$

i.e., $x_1 = \frac{C}{A}$ and $y_1 = -\frac{2Ba}{A}$

Illustration 65

Show that the locus of the poles of normal chords of $y^2 = 4ax$ is $(x + 2a)\ y^2 + 4a^3 = 0$.

Solution :

Given parabola $y^2 = 4ax$ ...(1)

Equation of normal of (1) is

$$y = mx - 2am - am^3 \quad ...(2)$$

Let (x_1, y_1) be the pole of (2) with respect to (1), then (2) is the polar (x_1, y_1) with respect to (1) i.e.,

$$yy_1 = 2a\ (x + x_1)$$

or $yy_1 = 2ax + 2ax_1$ ...(3)

Comparing (2) and (3) we get

$$\frac{y_1}{1} = \frac{2a}{m} = \frac{2ax_1}{-2am - am^3}$$

Then we get $y_1 = 2a/m$ ...(4)

and $x_1 = -2a - am^2$ ..(5)

Eliminating m between (4) and (5) we get

$$x_1 = -2a - a\left(\frac{2a}{y_1}\right)^2$$

$\Rightarrow$ $(x_1 + 2a)\, y_1^2 + 4a^3 = 0$

$\therefore$ Required locus of (x_1, y_1) is $(x + 2a)\, y^2 + 4a^3 = 0$

Illustration 66

Find the locus of the poles of the chords of the parabola $y^2 = 4ax$ which subtend a constant angle at the vertex.

Solution :

The given parabola is $y^2 = 4ax$ ...(1)

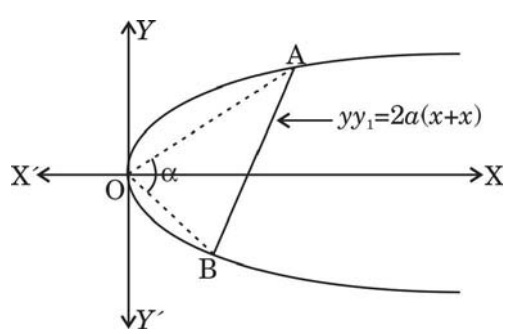
Let (x_1, y_1) be the pole of a chord of $y^2 = 4ax$

Then the equation of the chord of the polar of (x_1, y_1) with respect to (1) is

$$yy_1 = 2a\,(x + x_1)$$

or $$\frac{yy_1 - 2ax}{2ax_1} = 1 \quad ...(2)$$

For the equation of the lines joining the points of intersection of (1) and (2) with the vertex O(0, 0)



The making (1) homogeneous with the help of (2), we get

$$y^2 = 4ax\left(\frac{yy_1 - 2ax}{2ax_1}\right)$$

or $$4ax^2 - 2y_1xy + x_1y^2 = 0 \quad ...(3)$$

If chord subtends an angle α at the vertex then

$$\angle AOB = \alpha$$

$$\therefore \quad \tan\alpha = \frac{2\sqrt{h^2 - ab}}{|a+b|} = \frac{2\sqrt{(y_1^2 - 4ax_1)}}{|4a + x_1|}$$

or $$(4a + x_1)^2 \tan^2\alpha = 4(y_1^2 - 4ax_1)$$

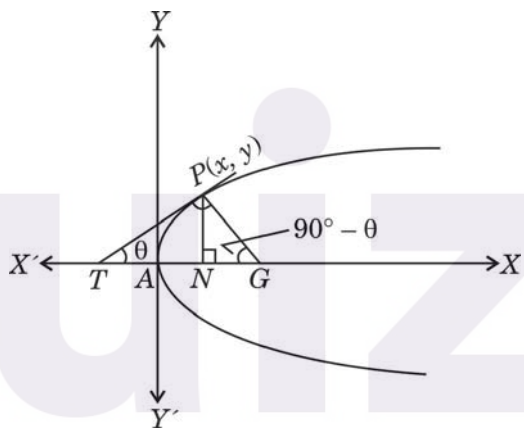
$\therefore$ The required locus of (x_1, y_1) is

$$(4a + x)^2 \tan^2\alpha = 4(y^2 - 4ax)$$

where α is constant.

10. LENGTHS OF TANGENT, SUBTANGENT, NORMAL AND SUBNORMAL

Let the parabola $y^2 = 4ax$. Let tangent and normal at P(x, y) meet the axis of parabola at T and G respectively and let tangent at P(x, y) makes angle θ with the +ve direction of x-axis.



Then PT = Length of Tangent

PG = Length of Normal

TN = Length of Subtangent

and NG = Length of Subnormal

If A(0, 0) is the vertex of the parabola

. PN = y

$\therefore$ PT = PN cosec θ = y cosec θ

PG = PN cosec (90 − θ) = y sec θ

TN = PN cot θ = y cot θ

and NG = PN cot (90 − θ) = y tan θ

where $\tan\theta = \frac{2a}{y} = m$ [slope of tangent at P(x, y)]

Illustration 67

Find the length of tangent, subtangent, normal and subnormal $ty^2 = 4ax$ **at** $(at^2, 2at)$.

Solution :

. Equation of tangent of $(at^2, 2at)$ of parabola $y^2 = 4ax$ is

$$ty = x + at^2$$

$$\text{Slope of this tangent } m = \frac{1}{t}$$

Let tangent makes angle θ with +ve direction of x-axis

$$\tan\theta = \frac{1}{t}$$

then $$t = \cot\theta$$

$$\text{Length of tangent at } (at^2, 2at) = 2at \operatorname{cosec}\theta$$

$$= 2at\sqrt{(1+\cot^2\theta)} = 2at\sqrt{(1+t^2)}$$

$$\text{Length of normals at } (at^2, 2at) = 2at \sec\theta$$

$$= 2at\sqrt{(1+\tan^2\theta)}$$

$$= 2a\sqrt{(t^2+t^2\tan^2\theta)} = 2a\sqrt{(t^2+1)}$$

$$\text{Length of subtangent at } (at^2, 2at) = 2at\cot\theta = 2at^2$$

$$\text{Length of subnormal at } (at^2, 2at) = 2at\tan\theta = 2a$$

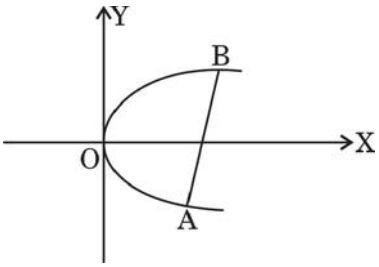
11. DIAMETER OF A PARABOLA

Def. Diameter of a conic is the locus of middle points of a series of its parallel chords.

Equation of diameter of a parabola :

Let the parabola be $y^2 = 4ax$ ...(1)

Let AB be one of the chords of a series of parallel chords having slope m.



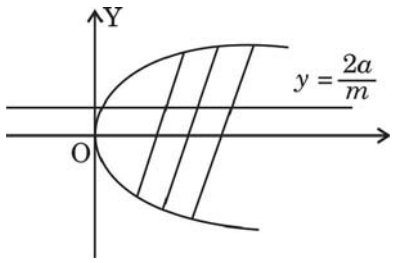
Let $P(\alpha, \beta)$ be the middle point of chord AB, then equation of AB will be $T = S_1$

or $$y\beta - 2a(x + \alpha) = \beta^2 - 4a\alpha \quad ...(2)$$

$$\text{Slope of line (2)} = \frac{2a}{\beta}$$

But slope of line (1) i.e., line AB is m.

$$\therefore \quad \frac{2a}{\beta} = m \text{ or } \beta = \frac{2a}{m}$$



Hence locus of P(α, β) i.e., equation of diameter (which is the locus of a series of parallel chords having slope m) is

$$\boxed{y = \frac{2a}{m}} \qquad ...(3)$$

Clearly line (3) is parallel to the axis of the parabola.

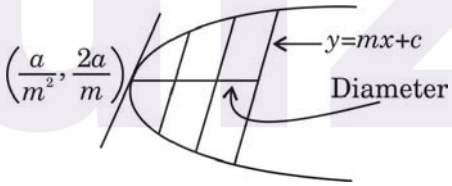
Thus a diameter of a parabola is parallel to its axis.

Note : **1.** The point is which any diameter meets the curve is called the extremity of the diameter.

2. Any line which is parallel to the axis of the parabola drawn through any point on the parabola is called diameter and its equation is y-co-ordinate of that point. If point on diameter (x_1, y_1), then diameter is $y = y_1$.

Remark 1. The tangent at the extremity of a diameter of a parabola is parallel to the system of chords it bisects.

Let y = mx + c (c variable) represents the system of parallel chords, then the equation of the diameter of



$$y^2 = 4ax$$

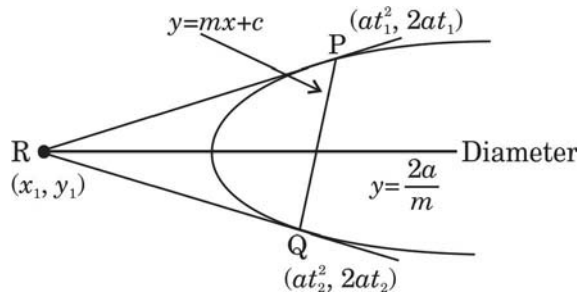
is $$y = \frac{2a}{m}$$

The diameter meets the parabola $y^2 = 4ax$ at $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ and tangent is $y = mx + \frac{a}{m}$ which is parallel to y = mx + c.

Remark 2 : Tangents at the end of any chord meet on the diameter which bisects the chords.

If extremities of the chord be $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ then its slope

$$m = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2} \Rightarrow m = \frac{2}{(t_2 + t_1)}$$



$\therefore$ Equation of diameter is

$y = 2a/m \Rightarrow y = a(t_1 + t_2)$ ...(1)

Now tangents at $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ meet at a point $[at_1t_2, a(t_1 + t_2)]$ which lies on (1).

Alternative Method : Let equation of any chord PQ be $y = mx + c$

If tangents at P and Q meet at $R(x_1, y_1)$ then PQ is the chord of contact with respect to $R(x_1, y_1)$.

$\therefore$ Equation of PQ is

$$yy_1 = 2x(x + x_1) \quad \text{or} \quad y = \frac{2a}{y_1}x + \frac{2ax_1}{y_1}$$

which is identical to $y = mx + c$

$$m = \frac{2a}{y_1} \quad \text{or} \quad y_1 = \frac{2a}{m}$$

Hence locus of $R(x_1, y_1)$ is $y = \frac{2a}{m}$ which bisects the chord PQ.

Note : The quantity $4a \operatorname{cosec}^2 \theta$ is called the parameter of the diameter AQ. It is equal to length of the chord which is parallel to AT and passes through the focus.

i.e.,
$$a \operatorname{cosec}^2 \theta = a(1 + \cot^2 \theta) = a + a \cot^2 \theta$$

$$= a + \frac{a}{m^2}$$

$$= a + VL = SP$$

but length of focal chord if $P(at^2, 2at)$ is { . S (a, 0)}

$$a\left(t + \frac{1}{t}\right)^2$$

$$\tan \theta = \frac{2at - 0}{at^2 - a} = \frac{2t}{t^2 - 1}$$

or
$$t - \frac{1}{t} = 2 \cot \theta$$

$$a\left(t+\frac{1}{t}\right)^2 = a\left\{\left(t-\frac{1}{t}\right)^2+4\right\} = a\{(2\cot\theta)^2+4\}$$

$$= 4a\,\text{cosec}^2\theta$$

$$= 4 \,.\, SP$$

Illustration 68

If the diameter through any point P of a parabola meets any chord in A and the tangent at the end of the chord meets the diameter in B and C, then prove that $PA^2 = PB \,.\, PC$.

Solution :

The equation of the parabola referred to the diameter through P and tangent at P as axes is

$$y^2 = 4\lambda x \qquad ...(1)$$

where $\lambda = a\,\text{cosec}^2\theta$

Let QR be any chord of the parabola (1) let the extremities Q and R be $(\lambda t_1^2, 2\lambda t_1)$ and $(\lambda t_2^2, 2\lambda t_2)$

Then the equation of QR is

$$y(t_1+t_2) - 2x - 2\lambda t_1 t_2 = 0 \qquad ...(2)$$

It meets the diameter through P i.e., x-axis or y = 0 then (2) reduces

$$0 - 2x - 2\lambda\, t_1 t_2 = 0$$

or $$x = -\lambda t_1 t_2 = PA$$

Now tangent at Q is $$t_1 y = x + \lambda t_1^2 \qquad ...(3)$$

It meets the diameter through P i.e., x-axis or y = 0 then (3) reduces

$$0 = x + \lambda t_1^2$$

or $$x = -\lambda t_1^2 = PB \qquad ...(4)$$

Similarly $$-\lambda t_2^2 = PC \qquad ...(5)$$

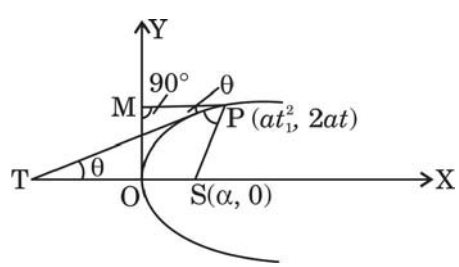
$\therefore$ $$(PA)^2 = (-\lambda t_1 t_2)^2 = \lambda^2 t_1^2 t_2^2$$

$$= (-\lambda t_1^2)(-\lambda t_2^2) \qquad \{\text{from equations (4) and (5)}\}$$

$$= PB \,.\, PC$$

12. THE OPTICAL PROPERTY OF A PARABOLA

The tangent at any point P on a parabola bisects the angle between the focal chord through P and the perpendicular from P on the directrx.



Sol. Let the parabola be

$$y^2 = 4ax \qquad ...(1)$$

Let $P(at^2, 2at)$ at a point on parabola (1) and let PM $\perp$ the directrix.

Equation of the tangent to parabola (1) at P is

$$yt = x + at^2 \qquad ...(2)$$

Slope of this tangent PT, $\tan\theta = \frac{1}{t}$ ...(3)

$$\text{Slope of PS} = \frac{2at}{a(t^2-1)} = \frac{2t}{t^2-1} = \frac{\frac{2}{t}}{1-\frac{1}{t^2}}$$

$$= \frac{2\tan\theta}{1-\tan^2\theta} = \tan 2\theta$$

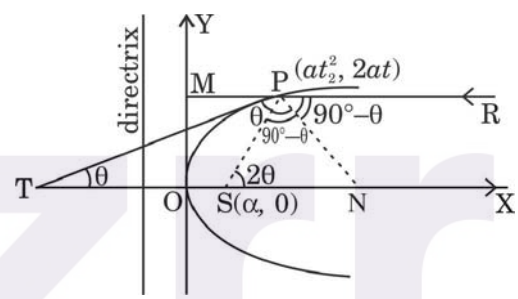
$\therefore$ $\angle PSX = 2\theta$

$\therefore$ $\angle TPS = \angle PSX - \angle PTS = 2\theta - \theta = \theta$

Hence $\angle SPT = \angle MPT = \theta$

Therefore PT bisects $\angle MPS$.

Note. Reflection from the parabola of a ray parallel to the axis :



In Δ PTN

$\angle TPN = 90$ and $\angle PTN = \theta$

Since PR || TN

$\therefore$ $\angle RPN = \angle TNP = 90 - \theta$

[. $\angle TNP = 90 - \theta$]

Also $\angle SPN = 180 - \angle PSN - \angle SNP$

$= 180 - 2\theta - (90 - \theta)$

$= 90 - \theta$

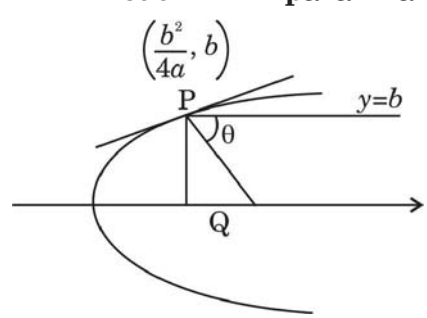
$\therefore$ $\angle SPN = \angle RPN$

Hence A ray parallel to the axis of the parabola after reflection from the parabola passes through the focus.

Illustration 69

A ray of light is coming along the line y = b from the positive direction of x-axis and strikes a concave mirror whose intersection with the xy-pane is a parabola $y^2 = 4ax$. Find the equation of the reflected ray and show that it passes through the focus of the parabola. Both a and b are positive.

Solution :



A ray of light along y = b intersects the parabola at $P\left(\frac{b^2}{4a}, b\right)$

Equation to the parabola is $y^2 = 4ax$

Differentiating (i) $\frac{dy}{dx} = \frac{2a}{y}$

Slope of normal at P is $-\frac{b}{2a}$

$\therefore$ If the angle between ray of light and the normal is θ then $\tan\theta = -\frac{b}{2a}$

Let the slope of PQ be m.

$$\tan\theta = \frac{-\frac{b}{2a} - m}{1 - \frac{b}{2a}m} \Rightarrow -\frac{b}{2a} = \frac{m + \frac{b}{2a}}{1 - \frac{b}{2a}m}$$

$$-\frac{b}{2a} + \frac{b^2}{4a^2} = m + \frac{b}{2a}$$

$$m\left(\frac{4a^2 - b^2}{4a^2}\right) = -\frac{2b}{2a} \Rightarrow m = -\frac{4ab}{4a^2 - b^2}$$

$\therefore$ Equation to the reflected ray is $y - b = -\frac{4ab}{4a^2 - b^2}\left(x - \frac{b^2}{4a}\right)$

$$y - b = -\frac{4ab}{4a^2 - b^2}\left(\frac{4ax - b^2}{4a}\right)$$

$$4abx + (4a^2 - b^2)\,y - 4a^2b + b^3 = 0$$

If we put x = a, y = 0, we get

$$4a^2b + (4a^2 - b^2)\,0 - 4a^2b = 0$$

$\therefore$ Reflected ray passes through the focus.